

Slim – core hole Drilling

Introduction, basis of design,
best practices and lesson
learned

TOMMY, HADI, SETO, SAHAT, SUBUR, ANDR, JEFRI, DAYAT, ADHINOER



Slim – core hole drilling

- Slim – core hole Introduction (Tommy)
 - Slim – core hole Objectives and Considerations
 - Slim – core hole Indonesia Historical, Past and Recently
 - Conventional Drilling vs Slim – core hole drilling
- Slim – core hole basis of design (Tommy)
 - Slim – core hole well design
 - Slim – core hole key casing design consideration
 - Slim – core hole key rig equipment
 - Slim – core hole Well control
- Slim – core hole Rig selection (Hadi)
 - Slim – core hole rig type
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 - Slim – core hole rig selection
 - Conventional rig vs Slim – core hole rig

Slim – core hole drilling

- Slim – core hole drilling program outline (Hadi)
 - Rig Mob, Rig up and Pre Spud
 - Conductor hole drilling and conductor setting and cementing
 - Surface hole drilling and surface casing setting and cementing
 - Production hole drilling and production casing setting and cementing
 - Core hole drilling and perforated liner setting
 - Open hole log
 - Completion test
- Slim – core hole drilling organization (Seto)
 - Rig Site
 - Office
- Slim – core hole HSE (Seto)
 - Key risks and the mitigation

Slim – core hole drilling

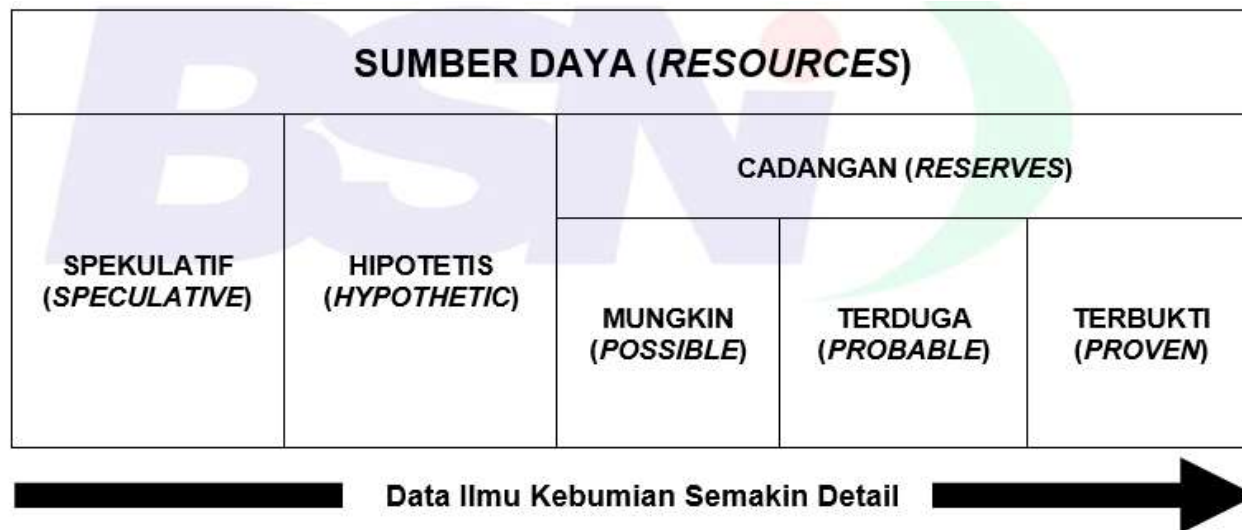
- Slim – core hole drilling operation (Sahat/Subur/Andri/Jefri)
 - Wellpad layout
 - Rig equipment line up
 - Coring operation
- Slim – core hole best practices and lesson learned (Sahat/Subur/Andri/Jefri)
 - Rotary drilling
 - Continuous coring
- Slim – core hole product knowledge (Dayat/Adhie)
 - Boart longyear history
 - Boart longyear coring rig
 - Boart longyear coring product
- Slim – core hole Opportunity for improvement (Team)

Slim – core hole Introduction

WHY Slim – core hole ??

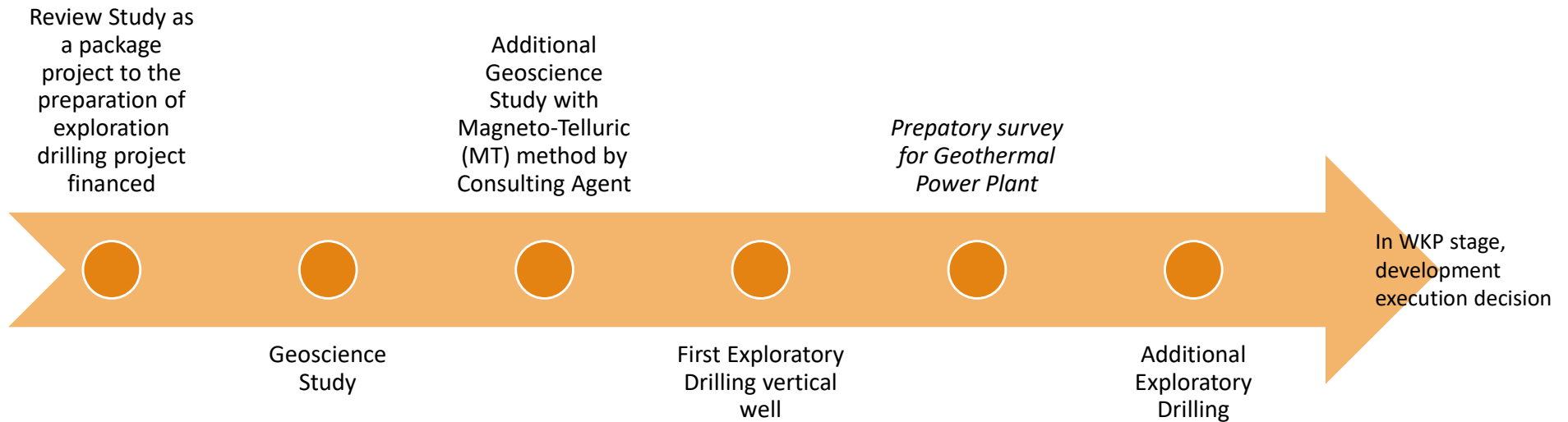
- To demonstrate the reserve
- To get as much sub surface information as feasible
- Answering the probable reserve in a timely and cost-effective manner
- Reduce the cost impact to get out of the concession

Geothermal Reserve (ref SNI 6009-2017)



Gambar 1 - Hubungan antara hasil kajian ilmu kebumian, sumber daya, dan cadangan energi panas bumi (modifikasi McKelvey, 1972)

Case example – Exploration preparation to execution



Slim – Core Hole Objectives and Considerations

Category	Definition	Measurement
PSPE Objectives	Determine Probable Reserve (ref SNI 6009-2017)	
	Obtaining geological, geochemistry, and geophysics survey data	Conceptual model of Geology, Geochemist, Geophysics, and integrated conceptual model
	Drill 1 (one) exploration well	Exploration well reach target depth as per conceptual model
	Able to prove geothermal conceptual model	Conceptual model validation before the drilling campaign preparation
	Able to estimate reservoir dimension	Reservoir size validation based on exploration well updated and impressive manifestation
	Able to determine reservoir rock characteristic	Core sample in reservoir section
	Able to determine reservoir fluid characteristic	- Self flowing well to determine reservoir fluid - Logging tool accessibility to bottom of the well (1-75" diameter K-10 Kuster)

Slim – Core Hole Objectives and Considerations

Category	Definition	Measurement
Company Objectives	Determine Probable Reserve (ref SNI 6009-2017)	Same with PSPE Objectives
	Confirming Probable Reserve Calculation can be used as reference to develop first 20 MW without doing more exploration	- No exploration stage requires - After IPB awarded, Company is able directly to drill 40 kg/s steam production well and 1 brine injector well with capacity of 120 MW and 1 condensate well with capacity of
	Confirming field development strategy	- Economic Calculation for each of the development strategy 2 x 10 MW for the first 3 years 2 x 20 MW for next 10 years
	- Exploration budget and timeline - Civil work duration and cost - Drilling time and cost - Well testing time and cost - Evaluation study time and cost	- Exploration total budget USD 14.4 M and time within 12 month - Civil work budget USD 7.5 M and within 6 months - Drilling work budget USD 5 M and within 4 months - Well testing budget USD 1 M and within 2 months - Evaluation study budget USD 0.5 M and within 6 months after well testing

Slim – Core Hole Objectives and Considerations

Category	Definition	Additional Objective
PSPE Objectives	Determine Probable Reserve (ref SNI 6009-2017)	• Confirming Probable Reserve Calculation can be used as reference to develop first 20 MW without doing more exploration • Confirming field development strategy • After IPB awarded, Company is able directly to drill 40 kg/s steam production well and 1 brine injector well with capacity of 120 MW and 1 condensate well with capacity of
	▪ Obtaining geological, geochemistry, and geophysics survey data	
	▪ Drill 1 (one) exploration well	
	▪ Able to prove geothermal conceptual model	
	▪ Able to estimate reservoir dimension	
	▪ Able to determine reservoir rock characteristic	
	▪ Able to determine reservoir fluid characteristic	

Ice Break Discussion

Exploration drilling target

- Exploration total budget USD 14.4 M and time within 12 month
 - Civil work budget USD 7.5 M and within 6 months
 - Drilling work budget USD 5 M and within 4 months
 - Well testing budget USD 1 M and within 2 months
 - Evaluation study budget USD 0.5 M and within 6 months after well testing



Innovation instead of Invention

Slim – core hole Indonesia Historical, Past and Recently

Sarulla ~1994 - 1997	Darajat 1996 - 1998	Ijen 2016 - 2017	Nage 2021	Huu – STM 2022	Toka Tindung 2022
• 9 exploration wells • 6 core hole wells • 5 deviated wells (3 Core deviated) • TD range 6662 ftMD – 8000 ftMD	• 7 exploration wells • All core hole wells • Slim – core hole wells (4-1/2" prod casing) • TD range 4402 ftMD – 7578 ftMD • 47 – 58 drilling days	• 1 and ½ exploration wells • Slim – core hole wells (4-1/2" prod csg) • TD 2000 mMD • 169 drilling days	• 1 Exploration well • Slim – core hole well (4-1/2" prod csg) • TD 1500 mMD • 225 drilling days	• 1 Exploration well • Slim – core hole well (4-1/2" prod csg) • TD 922.5 mMD • 193.3 drilling days	• 1 Exploration well • Slim – core hole well (4-1/2" prod csg) • TD 984.8 mMD • 204 drilling days

Darajat Slim – Core Hole 1997 – 1998

Ref : Riza et al, Proceeding 20th NZ Geothermal Workshop 1998

Well Name	TD (ft)	Core Interval Length (ft)	Top Resv	Max Record Temp deg F	Drilling Days	Hole Problem	Comment
S-1	4402	CHD 134 - 1445 HQ - 1978	Yes	423	51	None	
S-2	4730	CHD 134 - 1514 HQ - 1643	Yes	390	85	cement plugs (6 times) stuck pipe (9 days) one sidetrack left HQ pipe in hole	hole caving bridge off fishing
S-3	5087	CHD 134 - 1515 HQ - 2713	No	228	65	one cement plug	
S 4	5200	CHD 134 - 1410 HQ - 2871	No	275	52	None	
S-5	7467	HQ-4719	Yes	412	56	None	
S-6	6273	HQ-3348	No	300	58	stuck pipe (3 times) one sidetrack left HQ pipe in hole	encounter red clay
S-3A (deepening)	7578	HQ-2319 NQ - 173	Yes	350	35	stuck pipe (twice) HQ & NQ pipe left in hole	tight hole poor hole cleaning
S-4A (deepening)	7185	HQ-7926	Yes	304	93	two sidetracks stuck pipe (3 times) HQ pipe left in hole a lot of cement plugs	encounter red clay
DRJ-1A (deepening)	6282	HQ -2979	No	225	47	cement plugs (5 times)	encounter red clay hole caving

12. CONCLUSIONS

Over the almost two years of the slimhole program, six new wells were successfully drilled and cored, and three existing wells were deepened.

The program was very successful in obtaining the geological & reservoir information which it set out to achieve. Core analysis data and log interpretation yielded consistent and reliable information. The availability of core from the 6-1/8" hole sections in four of the wells was valuable in allowing direct comparison with the FMS logs obtained from the same section.

The 1996-98 Slimhole Drilling Program has demonstrated the ability of this type of drilling system to obtain quality information in a geothermal environment in a cost effective manner with minimal environmental impact. Some very useful drilling lessons and experience were gained from this program, which, when implemented on the next program, should result in reductions to drilling time and cost.

Disadvantages of Deep Slimhole Drilling

The main disadvantage of slimhole drilling is that the wells are generally not suitable for commercial geothermal production or injection, although in some fields such as the Geysers there is a move toward using a greater number of smaller injection wells, instead of a few larger ones.

The actual drilling time for a 1500 m deep continuously cored and subsequently hole opened slimhole (typically 60 - 65 days) is nearly twice that of a standard diameter geothermal production well of the same depth (28 - 33 days), depending on the type of rig in use and the experience of the contractor. This lower rate of penetration may be offset by the advantage of being able to continue drilling through lost circulation zones above the production casing shoe and the other factors noted above.

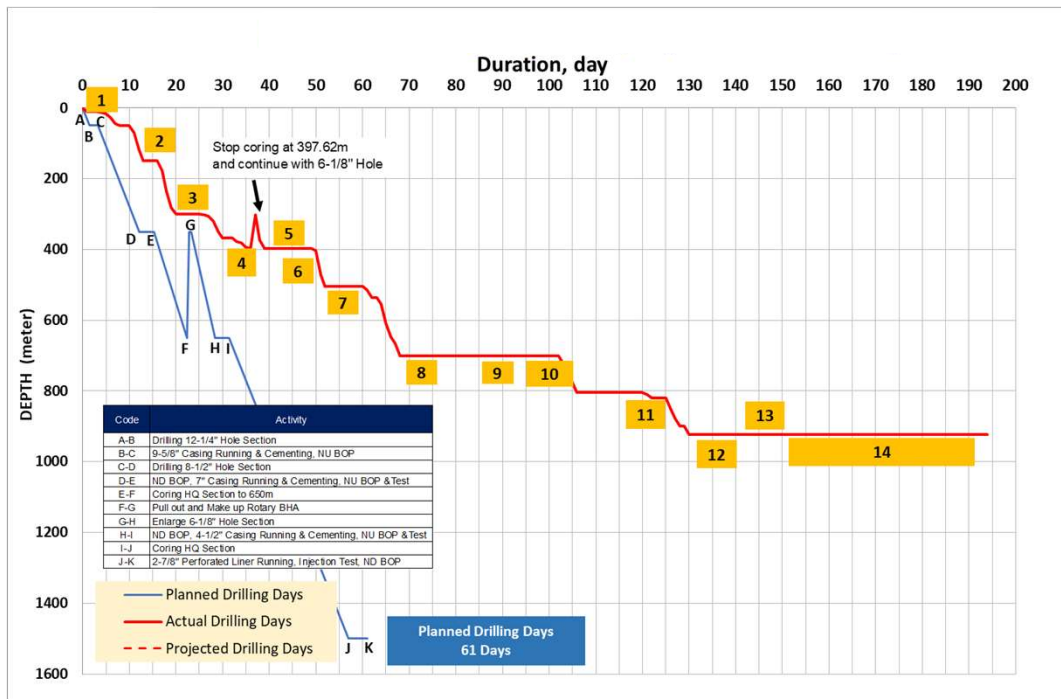
Cost

Depending on the location and rig availability, a 1500 m deep slimhole will typically cost US\$1.3-1.8 million. A standard diameter geothermal production well this deep might cost 3-4 times as much at about US\$5-6 million or more. In addition, the construction of roads, wellpads, and water supply all cost much less (typically 50 %) for a slimhole rig. Environmental permitting will also be quicker and cheaper in some jurisdictions.

What are your thoughts on
the time and cost figure?

is it trustworthy?

Slim – core hole Indonesia Historical, Past and Recently

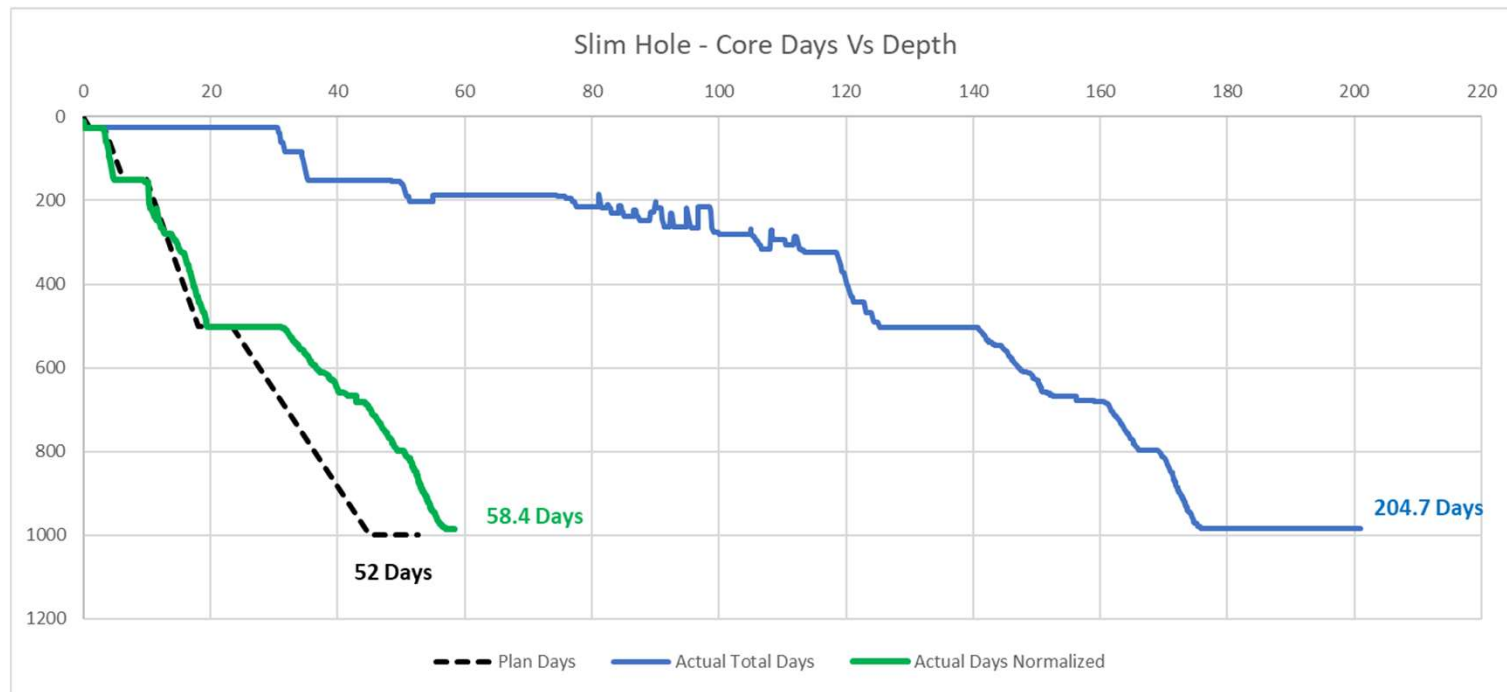


No.	Depth	Duration	Problem
1	10 m 13 m 16 m 28 m	± 7 days	A. Leaking on surface section. - Required to re-set riser and cement job - 3 times Grouting Job with cement B. Problem with Mud Pump
2	149.67m	± 3.5 days	Loss circulation starts at 143.6 m. Pumped LCM and Set 2 cement plugs
3	282.30m	± 3.3 days	Loss Circulation starts at 282.3m Pumped 2x Cement Plug
4	367.19m	± 2 days	Waiting on Thrust Bearing and TDS Crossover
5	397.62m	± 2 days	Hole Problem - Hole instability issue
6	397.62m	± 5 days	Waiting on Mud Pump Spare Part since 4th June 22 @ 12:00 hrs
7	504.89m	± 6.64 days	- Pull out of hole bit due to high torque - Incident on 15th Jun '22, back to operation on 21 June @21:30
8 & 10	701m	± 19 days	- Hard reaming during running in hole - Casing running stop at 372m and requirement to pull out casing - BHA trip to condition the hole - Cementing Plug stabilization
9	701m	11 days	Stop Operation due to community issue since 12th July 2022 @ 14:00 hrs until 23 July 2022 @ 14:00 hrs
11	805 m	5.5 days	Stop Operation due to waiting on spareparts (wireline winch drum) since 13th August 2022 @ 15:00 hrs.
12	922.65 m	on-going	NQ Pipe left in hole
13	922.65 m	10.75 days	Problem with Saver sub, waiting on equipment Pipe drop and fishing 2 times
14	922.65 m	33 days (on going)	Continue Fishing NQ Left in hole since 12th Sept

Total NPT = 109.8 Days

AFE Days : 61 Days
Actual Days : 193 Days
Normalized W/O NPT: 83.2 Days

Slim – core hole Indonesia Historical, Past and Recently



AFE Days : 52 Days
Actual Days : 204.7 Days
Normalized W/O NPT: 58.4 Days

Slim – core hole Indonesia Historical, Past and Recently

SURAT KEPUTUSAN **No. 189/SK/APMI/06/2017**

Pada hari Rabu, tanggal 7 Juni 2017, Rapat Pengurus APMI membahas Kebijakan dan situasi Industri Migas dan kaitannya dengan Standard Tarif Harian Operasi (THO) Rig On Shore.

- Melihat :
1. Tidak adanya perubahan THO yang berlaku sejak tahun 2015
 2. Meningkatnya standard kualitas, safety dan environment
 3. Meningkatnya biaya operasional dan perawatan
 4. Kenaikan Upah Minimum Regional dari tahun 2017/2018
 5. Harga sewa harian rig dunia naik 5 %*)
- Menimbang :
1. Hasil rapat pengurus APMI tanggal 7 Juni 2017
 2. Melihat kenyataan Tarif Harian Operasi yang sedang berjalan pada tahun 2017
 3. Kebutuhan Standard Tarif Harian Operasi (THO) guna mengurangi persaingan tidak sehat dan patokan bagi pemerintah dalam menentukan anggaran atau Owner (OE) tiap proyek
 4. Kesepakatan Konvensi Anggota APMI No. 074/KON/APMI/2017 tanggal 22 Februari 2017 sebagai hasil pertemuan Anggota APMI dalam rangka pembahasan standarisasi tarif sebagai tindak lanjut Kelompok Kerja (Pokja) antara SKK Migas, KKKS, dan APMI
- Memutuskan: Tarif Harian Operasi (THO) On Shore Rig APMI adalah:
1. Drilling : USD 19 – USD 23 Per Hari per HP
Work Over : USD 14 – USD 18 Per Hari per HP

Generic calculation

Coring Rig : 300 – 750 HP
Rig Cost : USD 5700 – 17250 /day
Spread cost : USD 15200 – 26750 /day

Slime – Core hole
Drilling Days : 60 – 65 days
Well cost : USD 0.9 MM – 1.7 MM

Conventional Drilling vs Slim – core hole drilling

CORING



ROTARY DRILLING



Conventional Drilling vs Slim – core hole drilling

CORING

- Polishing to cut the formation
- More to the rotation
- Low torque
- Less vibration
- High RPM
- Very low WOB
- Very fine cutting
- In gauge hole shape
- Coring head
- Circulation, High Pressure and low flow rate

ROTARY DRILLING

- Crush and Shear the formation
- More to the force applied, assisted by rotation
- Higher torque
- More vibrate
- relative low RPM
- Higher WOB
- Rough and bigger cutting
- Slightly over gauge
- Drill bit
- Circulation, High flow rate and relatively low pressure

Conventional Drilling vs Slim – core hole drilling

Slim- core hole

Slim Hole Well Design		Wellhead & Valve Stack	Well Control Equipment
Sub Surface Consideration	12-1/4" Hole 25 mMD/TVD 80 ftMD/TVD Conductor Casing 9-5/8", 40 ppf K55, BTC Surface Casing 7", 26 ppf, K-55, BTC Production Casing 4-1/2", 11.6 ppf K55, BTC HQ Pipe 3-7/8" (Oriented Core) 1000 mMD/mTVD 3280 ftMD	11", 2M	11" Diverter
Water Table 100-150 Mtrs 328 492	8-1/2" Hole 150 mMD/TVD 492 ftMD/TVD	7-1/6", 2 M	7-1/16" BOPE
Possible Fractures Target Zone 500 Mtrs 1640	6-1/4" Hole 500 mMD/mTVD 1640 ftMD/TVD	6" 600 master valve / 8" 900 refurbished valve	7-1/16" BOPE
1000 Mtrs 3280		7-1/6", 2M	

Big hole

Big Hole - Deviated		Wellhead & Valve Stack	Well Control Equipment
Sub Surface Consideration	36" Hole 37 mMD/TVD 120 ftMD/TVD Conductor Casing 30", 310 ppf ERW, Plain end Surface Casing 20", 106.5 ppf K-55, BTC Production Casing 13-3/8", 47 ppf L-80, BTC 10-3/4", 40.5 ppf K-55, BTC	29-1/2", 500 psi	29-1/2" Diverter
Water Table 160 Mtrs 524.8	26" Hole 160 mMD/TVD 525 ftMD/TVD KOP @ 300 mMD/TVD	21-1/4", 3M	21-1/4", 3M BOPE
Possible Fractures Target Zone 600 Mtrs 1968	17-1/2" Hole 600 mMD/580mTVD 1968 ftMD/1903ftTVD EOB @ 600 mMD/580mTVD 1968 ftMD/1903 ftTVD	12" 600 master valve	21-1/4", 3M BOPE
1400 Mtrs 4592	12-1/4" Hole 1400 mMD/ 1273 mTVD 4592 ftMD/4175 ftTVD	13-5/8", 2M	

Slim – core hole basis of design

WHAT is Slim – Core hole casing configuration?

- Conductor casing
- Surface Casing
- Production casing
- Perforated liner

Directional: Vertical

EVAL PROG		HOLE SIZE		BOPE	
80 ft MD/TVD		14-3/4" Hole		BOPE	
		10-3/4" 40.5ppf H40 Conductor Casing		TBD	
410 ft MD/TVD		9-7/8" Hole		Unweighted WBM	
		7-5/8" 24ppf H40 BTC Prod. Casing		TBD	
1148 ft MD/TVD		PQ x 6-3/4" Hole		Unweighted WBM	
		3-1/2" Top of Liner		1057 ft MD	
		5-1/2" 15.5ppf K55 BTC Prod. Casing		TBD	
Wireline Logging: Weatherford Compact Microimager		PQ - 4.815" Hole (Oriented Core)		FW / Polymer Sweeps	
Well Testing: Multi-rate Injection PTS Logging Fall-off Test Flow Test - Self Flow, Compress w/Air, DH Sampling if unable to flow		1st Feed Zone - 1300' - 1650' TVD			
2625 ft MD/TVD		2nd Feed Zone - 2130' - TD		3-1/2" 9.2ppf J55 Production Liner - Slotted	
				BHT: 358 F	
Well Schematic					

Slim – core hole well design

WELL OBJECTIVES

- ✓ Drill Vertical well to 1000 Meters
- ✓ Continuous core sample from below surface casing to projected Top of Reservoir and Fracture Orientation prediction
- ✓ Continuous core sample in reservoir section to TD and Fracture Orientation prediction
- ✓ To enable a PT to be conducted in the well , Minimum Tool ID is ~2.25", clearance ~3"
- ✓ Well design for self flow – not expected however

GIVEN

- ✓ Spud date Q3 2021
- ✓ Desco SP8500SA Rig
- ✓ Slim hole – continuous core well

CASING DESIGN COMPLEXITY

- ✓ Production casing size – Optimum diameter vs Market availability
- ✓ Reservoir coring size – Maximum hole clearance vs Time Performance

Slim – core hole well design

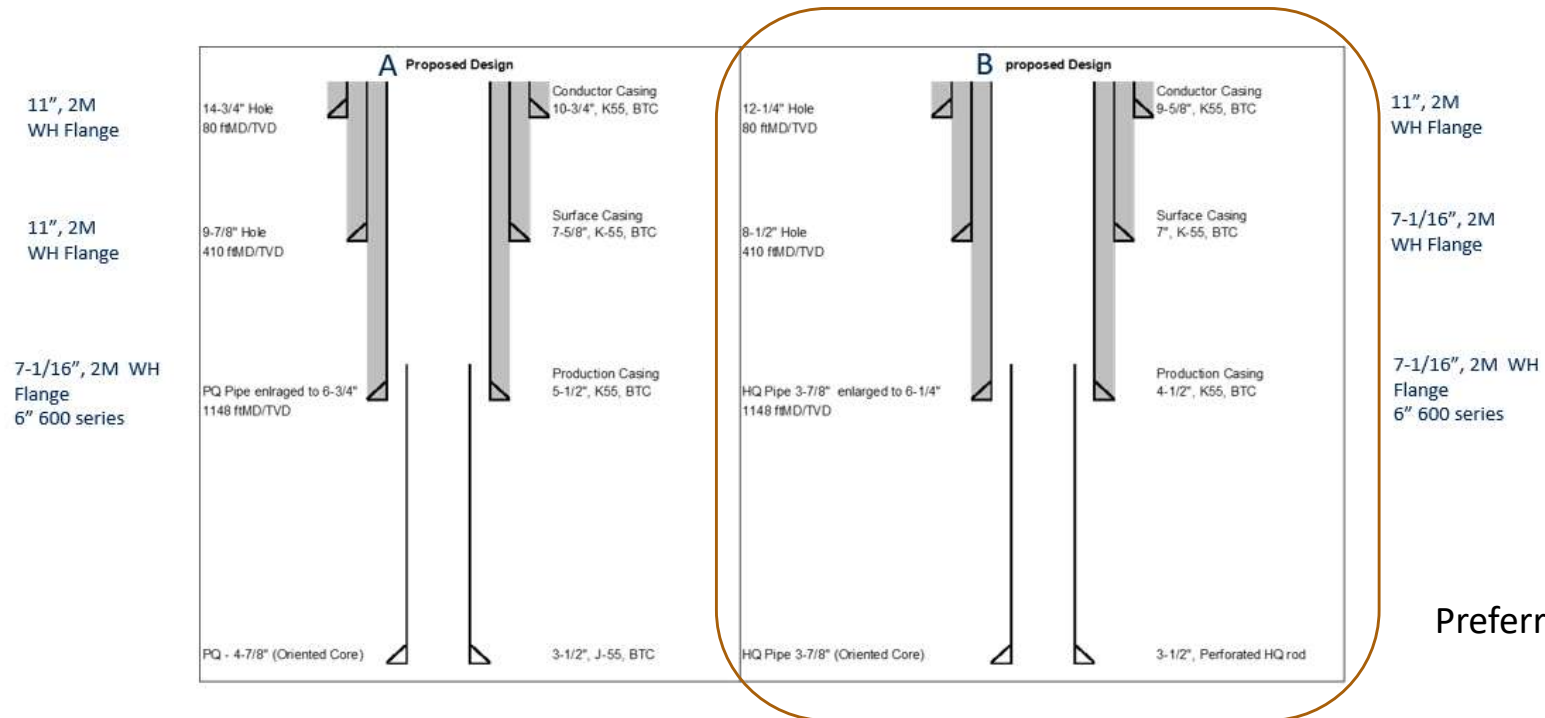
A Proposed Design Remarks	Check list	Bitung Well Objectives	Check list	B proposed Design Remarks
PQ Rate Maximum Depth 1500 m	Y	Drill Vertical well to 1000 Meters	Y	NQ Rate Maximum Depth 1500 m
	Y	Continuous core sample from below surface casing to projected Top of Reservoir and Fracture Orientation prediction	Y	
	Y	Continuous core sample in reservoir section to TD and Fracture Orientation prediction	Y	
PQ Rod Body Diameter 4"	Y	to enable a PT to be conducted in the well , Minimum Tool ID is 2.25 Inch	Y	HQ Mid Body and Inner joint Diameter 3.06 "
Cemented Production Casing with 600 series (1271 psi derated) and 2M wellhead (1905 psi derated)Valves system Fluid Temperature prediction T81 deg C, Max WHP 991 psi	Y	Well design for self flow	Y	Cemented Production Casing with 600 series (1271 psi derated) and 2M wellhead (1905 psi derated)Valves system Fluid Temperature prediction T81 deg C, Max WHP 991 psi

Generate and Select Alternative (s)

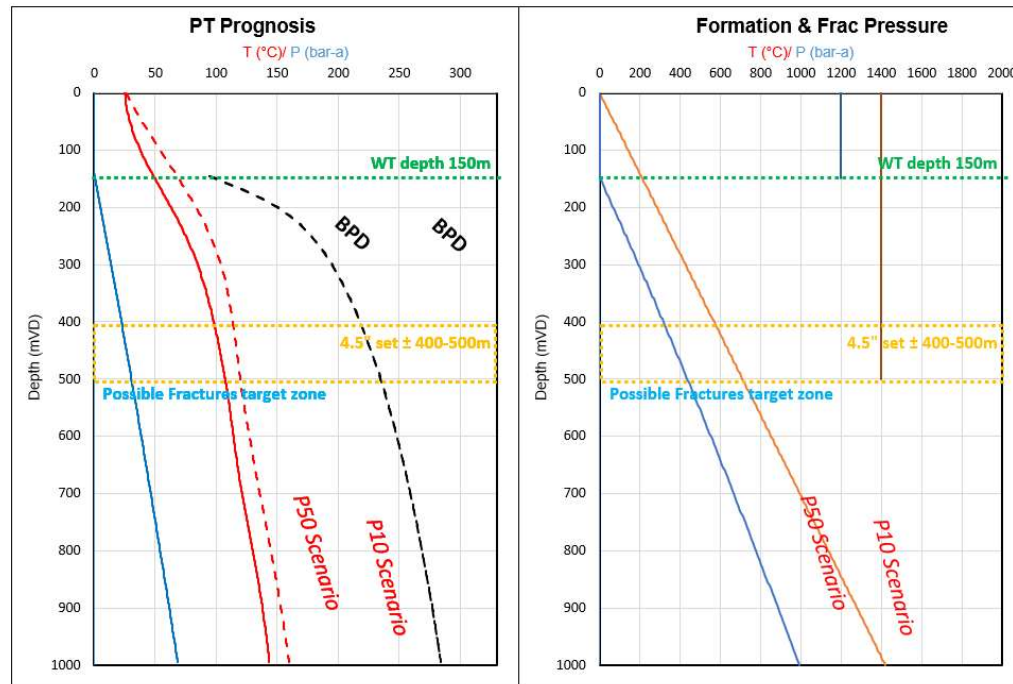
Alternative trade off

A Proposed Design		B proposed Design	
Plus	Bigger Production casing size	Delta	Smaller production casing size
	More option for coring in reservoir section (PQ and NQ) *		Less option for coring in reservoir section (PQ)*
	More wellbore clearance for fracture orientation with logging tool (FMI, etc)		Tigh clearance for fracture orientation with logging tool (FMI, etc)
Delta	7-5/8" and 5-1/2" casing are not common stock for Geothermal in Indonesia - Possible for longer delivery time	Plus	All casing sizes are common stock for Indonesia geothermal - Possible for more faster delivery and some stock available in Country
Remarks	To address additional time to handling PQ core barrel, can be considered to coring with HQ for production hole section		Possible for faster coring time

Slim – core hole well design



Casing Design key Consideration



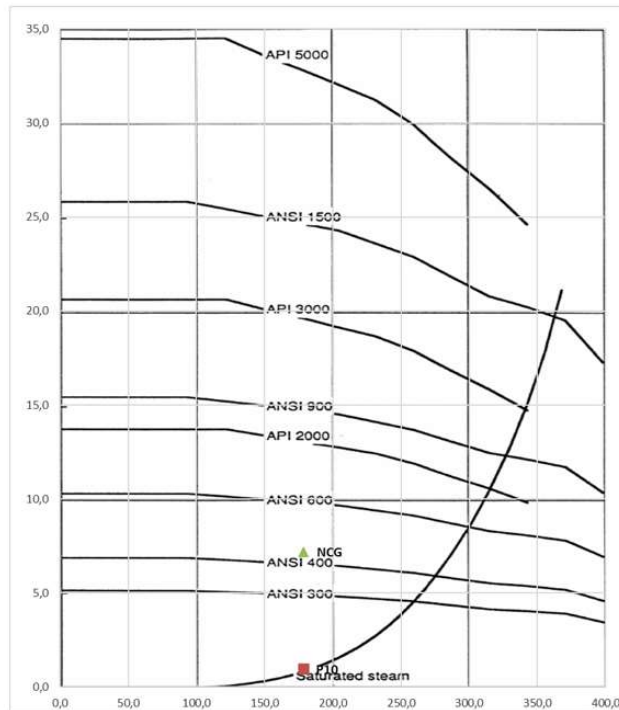
Predicted
Formation
Pressure – Sub
Normal
Pressure.

Surface Casing
cover predicted
ground water
table.

Production
Casing in the
range +/- 400 –
500 m

Casing Design key Consideration

Wellhead and BOPE



Hole Section	BOP Size	Wellhead	MAWHP	Minimum BOP Test Pressure
12-1/4"	11" 500 psi Diverter	Temporary Wellhead 11", 3M threaded onto 9-5/8" casing	59	Diverter pressure test will conduct at machine shop
8-1/2"	7-1/16" 3M BOPE	Temporary Wellhead 7-1/6", 3M threaded onto 7" casing	487	600
6-1/4"	7-1/16" 3M BOPE	DSA x 8"900 Master Valve x DSA x 7-1/16 2M Casing Head Flange threaded onto 4-1/2" casing	1042	1200

BOP Testing

- BOP maximum rating working pressure test at the stump
- BOP pressure test to MASP + 100 psi safety factor
- BOP pressure test should not exceed 70% of API Burst Rating of weakest exposed wellbore component

Minimum Specification for Wellhead and Master Valve

- Master Valve : ANSI 600
- Wellhead Equipment & Flange : 2000 psi pressure rating

Casing Design key Consideration

Master valve

Maximum Design Pressure : 1042 PSI with 353 deg F

✓ Refer to the updated prognosis in June 2021

✓ Boiling Point Temperature and Pressure at TD

Tab. 2: Temperature rating for wellhead equipment as per API 6A (2011), Annex G.

Classification	Operating Temperature range	
	(°F)	(°F)
X	-18 to 180	0 – to 350
Y	-18 to 650	0 to 650

Tab. 3: Pressure temperature ratings for 6B flanges as per API 6D, Annex G.

Pressure rating for classes K to U psi	Derated pressure	
	Class X (psi)	Class Y (psi)
2000	1905	1430
3000	2860	2145
5000	4765	3575

Tab 4: Working Pressure vs temperature for API SPEC 6D Geothermal Mater valves 1.1 material (A515 Gr70) as per ASME 16-34, 2013.

Temperature (°F)	Working pressure (psi)				
	ANSI 150	ANSI 300	ANSI 600	ANSI 900	ANSI 1500
-20 to 100	284	741	1481	2222	3703
212	257	676	1352	2028	3379
302	229	654	1308	1961	3269
392	200	635	1271	1906	3176
482	175	608	1217	1825	3041
572	148	577	1155	1733	2888
617	135	561	1123	1684	2808
662	122	545	1089	1635	2724

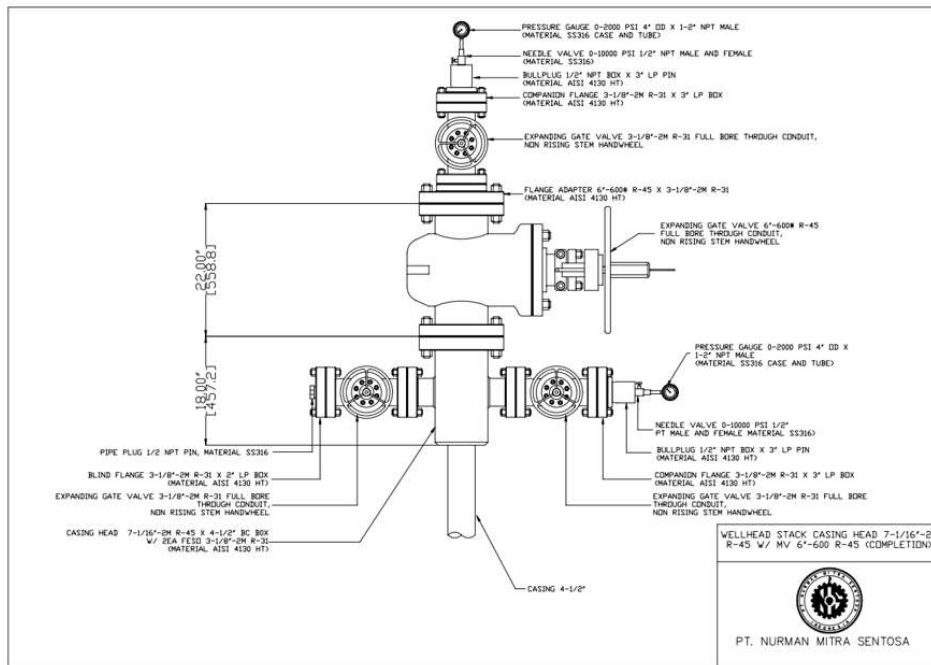
Minimum Specification for Wellhead and Master Valve

Wellhead Equipment & Flange : 2000 psi pressure rating

Master Valve : ANSI 600

Casing Design key Consideration

Wellhead Stack



4 Brand new 6" ANSI 600 valves are prepared for additional well after 2nd well and as contingency to replace 8" ANSI 800 refurbished valve if needed.

Wellhead 2000 psi rating :

- ✓ Wellhead flange 7-1/16", 2M with 4-1/2"BTC bottom connection
- ✓ Side outlet flange 3-1/8", 2M

The valves 2000 psi rating :

- ✓ 6"-ANSI 600
- ✓ 3-1/8" ANSI 600

NZS 2403:2015 PRODUCTION CASING DESIGN ANALYSIS

Description	Casing OD	Casing Weight	Casing Grade	ID	Yield Strength	Tensile Strength
4-1/2" Surface Casing	4.5	11.7	K55	4	55000	95000
2.10.2 Assessing Axial Loading before and during cementing	Calculated	Minimum Design Factor				
Maximum Axial Load						
Axial SF	9.57	Pass > 1.8				
Temperature Derated Axial SF	8.61	Pass > 1.8				
Bending Force						
Axial SF	18.69	Pass > 1.8				
Temperature Derated Axial SF	16.82	Pass > 1.8				
Force Resultant						
Axial SF	6.33	Pass > 1.8				
Temperature Derated Axial SF	5.70	Pass > 1.8				
2.10.3 - Axial Loading after Cementing	Calculated	Minimum Design Factor				
Hot Production Load						
Pipe Axial Yield SF	0.73	If < 1.4, design for yield - see MHA below				
Temp Derated Axial SF	0.66					
2.10.3.6 - Fluid Lifting Force on Anchor Casing	Calculated	Minimum Design Factor				
F _W (Lifting Force)	14.02	Pass > 1.8				
Axial SF	12.61	Pass > 1.8				
Temperature Derated Axial SF						
2.10.6.2 Hoop Stress - Internal Yield Burst Cementing	Calculated	Minimum Design Factor				
Burst Differential Pressure						
Pipe Burst SF	4.69	Pass > 1.5				
Temp Derated Pipe Burst SF	4.22	Pass > 1.5				
2.10.6.3 Hoop Stress - Internal Yield Burst Max Well Pressure (Steam or Gas)	Calculated	Minimum Design Factor				
Max Well Pressure						
Pipe Burst SF	5.13	Pass > 1.8				
Temp Derated Pipe Burst SF	4.62	Pass > 1.8				
2.10.6.4 Hoop Stress - Biaxial Wellhead Load	Calculated	Minimum Design Factor				
W _H						
Biaxial W _H Load SF	5.90	Pass > 1.5				
Biaxial W _H Load SF Temp Derat	5.31	Pass > 1.5				
2.10.7.2 Hoop Stress - Collapse Cementing	Calculated	Minimum Design Factor				
Collapse Differential Pressure						
Pipe Burst SF	7.75	Pass > 1.5				
Temp Derated Pipe Burst SF	7.19	Pass > 1.5				
2.10.7.3 Hoop Stress - Production Collapse or Lost Returns Drilling	Calculated	Minimum Design Factor				
Evacuated Collapse						
Pipe Burst SF	6.92					
Temp Derated Pipe Burst SF	6.42					
Collapse Trapped Fluid - Outer Casing Burst Pressure						
API Collapse SF	1.94	Pass > 1.2				
Derated Collapse SF	1.80	Pass > 1.2				
Collapse Trapped Fluid - Ductile Rupture of Outer Casing						
API Collapse SF	0.48	Pass > 1.2				
Derated Collapse SF	0.45	Pass > 1.2				
2.10.3.4 - Axial Loading after Cementing - Modified Holiday Approach	Calculated	Minimum Design Factor				
Uniaxial MHA SF	1.32					
Hoop Stress - Pressure Test	Calculated	Minimum Design Factor				
Pipe Burst SF	4.40	Pass > 1.8				
Temp Derated Pipe Burst SF	4.01	Pass > 1.8				

- ✓ 4-1/2" production casing satisfied NZS 2403 minimum design factor except for the trapped fluid in the annulus of 7" casing x 4-1/2" when the well heat up.
- ✓ Cement quality and integrity behind 4-1/2" is critical to prevent the trapped fluid

Slim – core hole key rig equipment

SP8500SA

VERTICAL & ANGLE DRILLING, CRAWLER MOUNTED
FOR EXPLORATION DRILLING (DIAMOND CORE & DTH DRILLING)

400 HP Coring rig



DIMENSIONS & WEIGHT

Length-transport	11.5M
Width	2.3M
Height-transport	3.1M
Weight	21.5ton

ENGINE

Model	Doosan PU126Ti
Rated power	400HP
Fuel capacity	400 liter
Cooling system	water cooling

DRILL HEAD & ROTATION

The Rotary Head Sliding type to easily handle drill pipe and casing	
Bullshaft thread	3-1/2 API IF, NQ, HQ, PQ
Performance	two motors with 3 stages Eaton 6-490 & Rexroth 125
Max. RPM / Max. Torque	1ST : 0~120 / 750kg.M 2ND : 200~700 / 280kg.M 3RD : 700~1000 / 220kg.M

FEED SYSTEM

Traverse cylinder	200mm*140mm*3750mm
Head travel	7.5M
Pull-up capacity	45ton
Pull-down capacity	20ton

HYDRAULIC SYSTEM

General	4 pump open loop system
Pump model	TSV200*2+K3V112DP
Fuel capacity	400liter
Operating pressure	280kg/cm2

CONTROL PANEL

All controls gauges are grouped at the operator's station within sight of the borehole
1. Install panel cover & chair for operator

LIGHT & ELECTRICAL SYSTEM

Mast	2-60 watt floodlights
Control panel	2-gauge floodlights
Work	2-60 watt floodlights

WORKING TABLE & HYD. CLAMP

Max. Diameter	11"
Double clamp	11"

DRILL PIPE & CASING

Drill pipe length	Max. 6.0M
Max. diameter	11"

HYD. OUTRIGGERS

Four hyd. Jacks & outriggers	
Ground clearance	600mm
Width	3000mm

DRILL PIPE BREAKOUT

Cylinder operated pipe wrench	
Torque break-out	6.0 tons

WINCH

Winch	30M / 10ton (Dia. 16mm)
Wire line	Drum for 2500M / 1000kg Spooler for wire - 6.0mm

CRAWLER BASE

Crawler model	Doosan Solar 75 - TM22VC
Crawler type	Steel track
Crawler speed	2.5 km/h

MUD PUMP

Triplex piston pump from American Manufacturing	
Model	AW1122BCD
Capacity	200LPM @ 70bar

AUTO ROD HANDLER

Drill Rod	NQ, HQ & PQ X 6.0M
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- 2 -

Drill Head & Rotation

Max RPM / Max Torque	RPM	Torque	
		Kg.M	Lb.ft
1 st	0-120	750	5424.76
2 nd	200-700	280	2025.24
3 rd	700-1000	220	1591.26

Feed System

Head Travel	7.5 M	24.6 ft
Pull-up Capacity	45 ton	88,185 Lbs
Pull-down Capacity	20ton	44,092 Lbs

Working Table & Hyd.Clamp

Drill Pipe length	6M	19.69 ft
Maximum		11 inch

Drill pipe Breakout

Torque break-out	6tons	18,515,760,195	ftlb
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Slim – core hole key rig equipment

Hoisting

Hoisting	Maximum load	Conductor	Surface Hole	Production
	Depth	120 ft	490 ft	1642 ft
	Drilling	9.7 Klb	18.1 Klb	7.4 Klb
	Casing	4.8 Klb	13.7 Klb	19.2 Klb

VS

Feed System			
Head Travel	7.5 M	24.6	ft
Pull-up Capacity	45 ton	88,185	Lbs
Pull-down Capacity	20ton	44,092	Lbs

Adequate for drilling and coring



Slim – core hole key rig equipment

Circulation

Circulation	Hole Size	Q @Va 150 ft/Hr	Q @Va 200 ft/Hr
		GPM	GPM
	12-1/4"	844	1125
	8-1/2"	368	490
	6-1/4"	165	219
	3-7/8"	2	2

VS

MUD PUMP

Triplex piston pump from American Manufacturing

Model AW1122BCD

Capacity 200LPM @ 70bar

Flow 52.8 GPM @ 1015 PSI

Adequate for Coring only

Slim – core hole key rig equipment

Circulation

Circulation	Hole Size	Q @Va 150 ft/Hr	Q @Va 200 ft/Hr
		GPM	GPM
	12-1/4"	844	1125
	8-1/2"	368	490
	6-1/4"	165	219
	3-7/8"	2	2

VS

PZJ (PZ-9) Drilling Service Ratings									
Piston Cylinder		Displacement				Maximum PSI		Pump RPM	JackShaft RPM
Diameter		Per Revolution		Per Minute					
Inch	mm	Gal.	Lit.	Gal.	Lit.	PSI	Kg/sq.cm2	130	582
7	178	4.50	17.03	585	2,214	2,639	186		
6.5	165	3.88	14.69	504	1,908	3,060	215		
6.25	159	3.59	13.59	466	1,764	3,310	233		
6	152	3.30	12.49	430	1,628	3,592	253		
5.5	140	2.78	10.52	361	1,367	4,274	301		
5	127	2.29	8.67	298	1,128	5,000	352		
4.5	114	1.86	7.04	242	916	5,000	352		
4	102	1.47	5.56	191	725	5,000	352		

1 pump adequate for most of the hole sizes, except for the Conductor

Slim – core hole key rig equipment

Rotation

Rotation	Hole Size	Depth	RPM range	Actual Torque range klbft
	12-1/4"	120 ft	30 -60	0.3 - 1
	8-1/2"	490 ft	30 - 80	0.5 - 1.5
	6-1/4"	1642 ft	30 - 70	0.2 - 1.7
	3-7/8"	2883 ft	200 - 350	0.2 - 1.5
	2-7/8"	3233 ft	250 - 380	0.25 - 1.2

VS

Drill Head & Rotation			
Max RPM / Max Torque	RPM	Kg.M	Lb.ft
1 st	0-120	750	5424.76
2 nd	200-700	280	2025.24
3 rd	700-1000	220	1591.26
Drill pipe Breakout			
Torque break-out	6tons	18,515,760,195	ftlb

The torque limit is still able to exceed the actual torque

Slim – core hole key rig equipment

Other consideration

- Drill pipe and drill string component : R1 instead of R3 with the size 3-1/2"
- Sub Structure clearance : BOPE stack high, flow line slope, the tank and solid control elevation
- Mud circulation system : Hivis mixing independently, Shale shaker mesh for drilling and coring cuttings
- Casing running and connection torque up : R1 casing, adequate space for big casing torque up and running, and perforated liner running tool
- Drilling parameter display and record : Torque reading and the depth counter

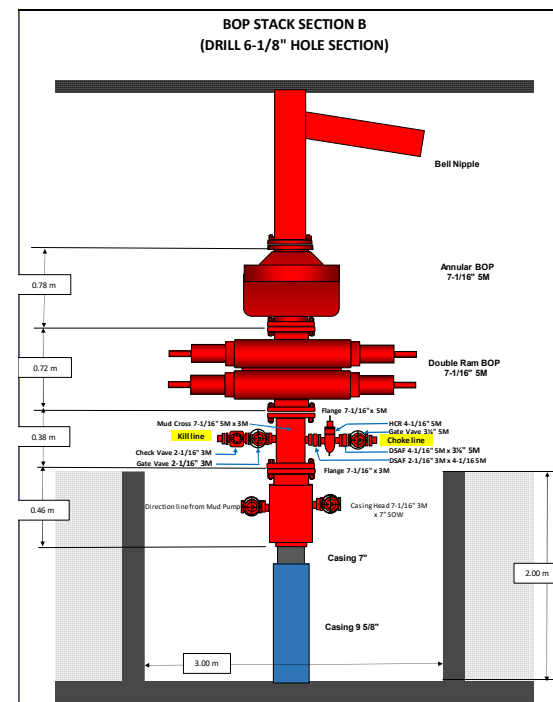
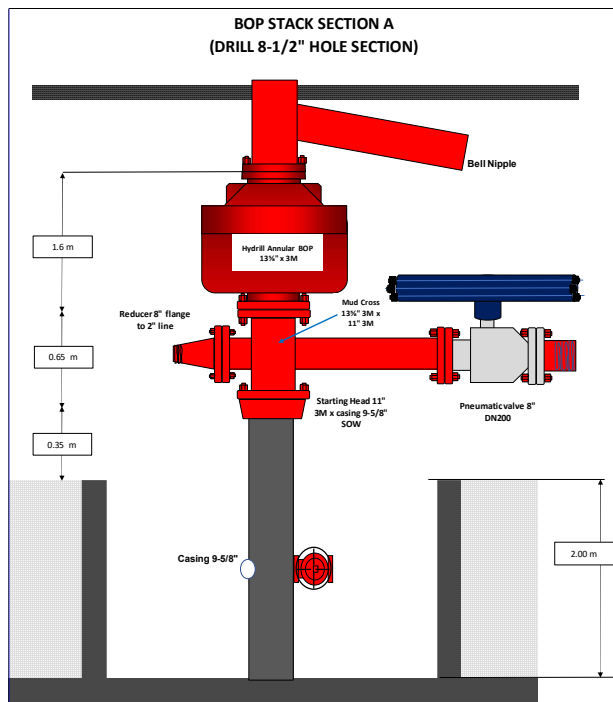
Slim – core hole Well control

What is Slim – core hole hazard??

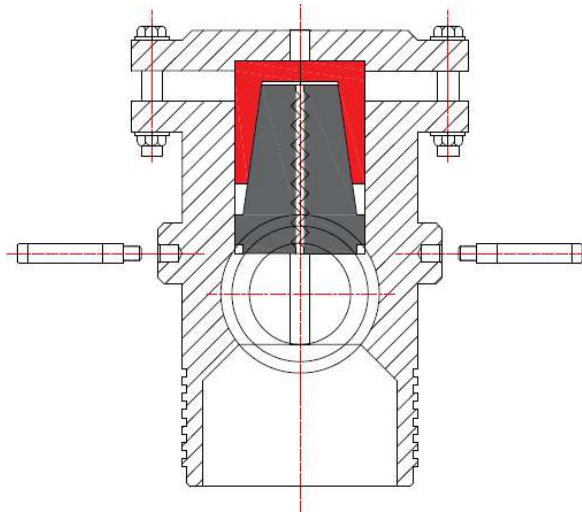
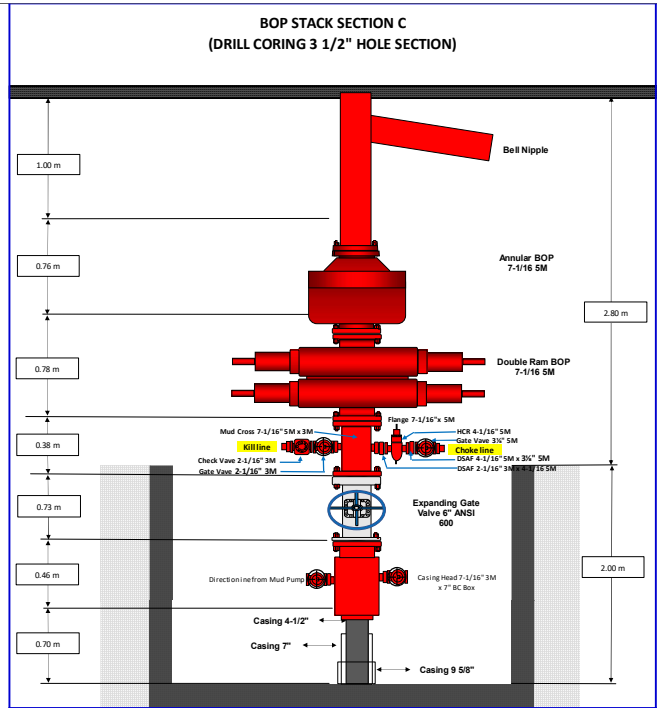
- Shallow gas
- Shallow hot water and steam
- Geothermal gasses
- Steam and hot water

Slim – core hole Well control

Rotary Drilling

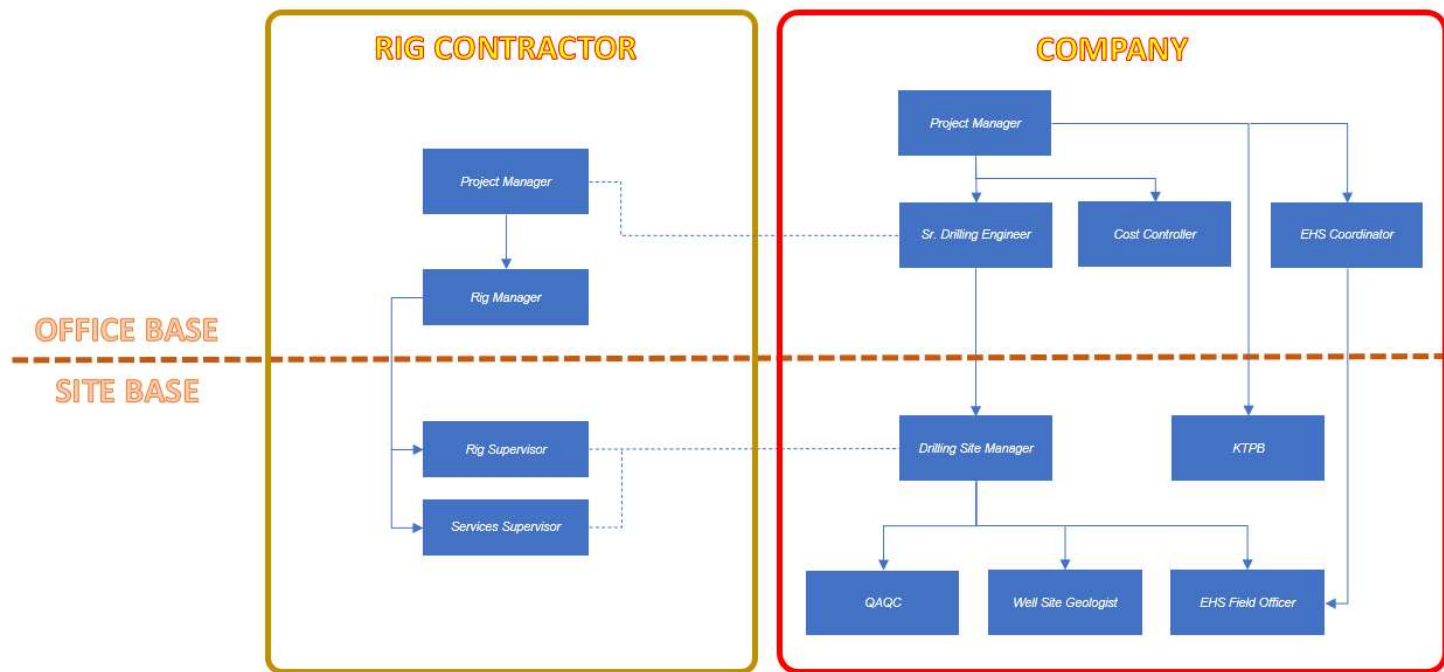


Coring



Well Control Barrier for Retrieving Core Barrel

Slim – core hole drilling organization



Slim – core hole key risk and mitigation

Key Risk

- Safety Commitment
- Contractor selection process
- Safety Management System
- Tools, equipment and people

Slim – core hole key risk and mitigation

Mitigation

- Contractor selection must fit for purpose with proven slim hole coring experience
- Develop bridging document between contractor and company
- Conduct Incident Free Operation workshop prior kick-off campaign
 - Vision and Mission of Project
 - Safe Work Practices
 - Hand Injury Prevention
 - Behaviour Based Safety Observation
 - Drop Object Prevention
 - Home Made Tools

Slim – core hole drilling operation

Wellpad layout



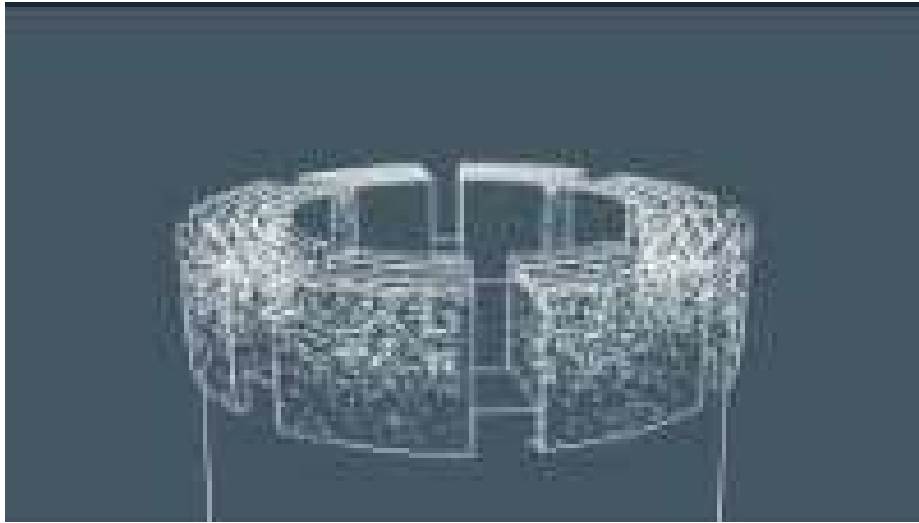
1. Rig Area
2. Water and mud pond
3. Mudpit and water treatment
4. Cementing area
5. Storage Area
6. Welding shop
7. Site office, site camp, and meeting room
8. Medic room and security pos
9. Fuel tank

Slim – core hole drilling operation

Rig layout



1. Rig floor
2. Mast
3. Possum belly, shaker, sand trap
4. Active tank
5. Mixing tank
6. Cooling tower
7. Accumulator
8. Generator
9. Triplex mud pump
10. FMC pump
11. Pipe and core skate



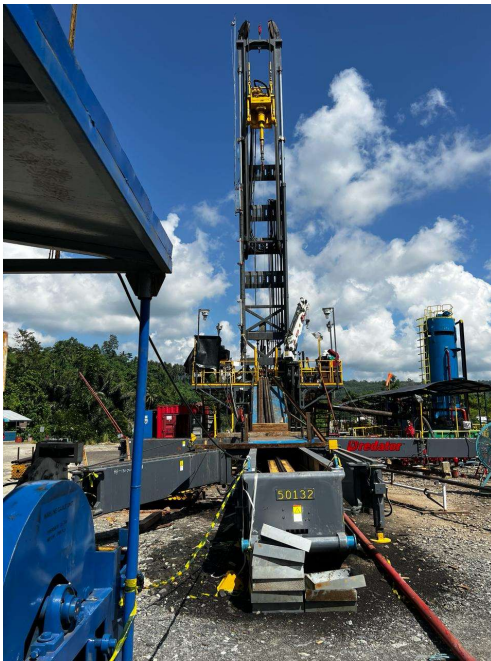
Coring System



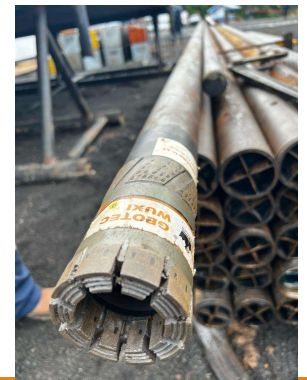
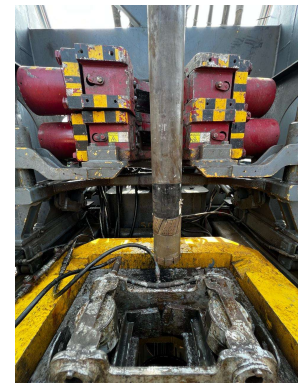
Slim – core hole

Best practices and lesson learned

➤ Preparation



Equipment Set Up



Slim – core hole

Best practices and lesson learned

➤ Execution



Slim – core hole

Best practices and lesson learned

➤ Execution



Slim – core hole

Best practices and lesson learned

➤ Execution



Slim – core hole

Best practices and lesson learned

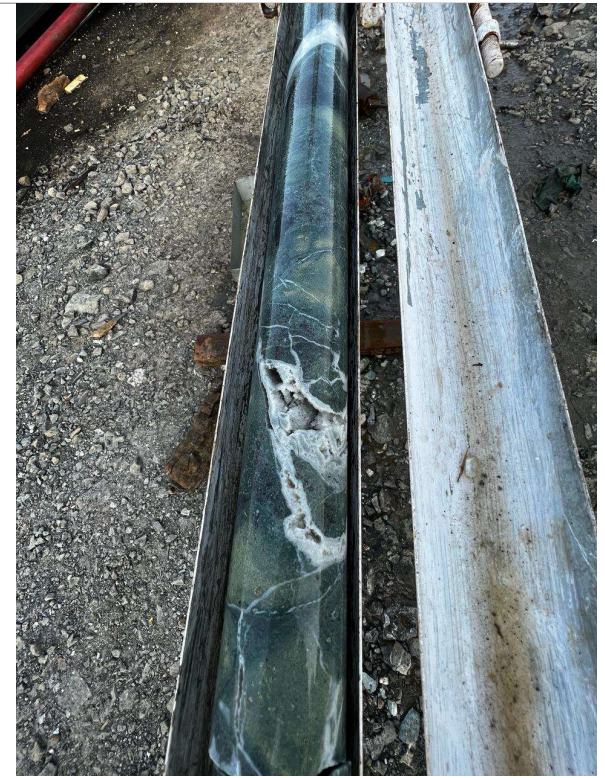
➤ Execution



Slim – core hole

Best practices and lesson learned

➤ Execution



Slim – core hole

Best practices and lesson learned

➤ Execution



Slim – core hole

Best practices and lesson learned

➤ Mitigation



Slim – core hole

Best practices and lesson learned

➤ Mitigation



Slim – core hole basis of design

Best practices and lesson learned

It can be grouped into

➤ Rotary Drilling

- Conductor hole drilling, conductor casing running and cementing
- Surface hole drilling, surface casing running and cementing
- Production hole drilling, production casing running and cementing

➤ Continuous coring

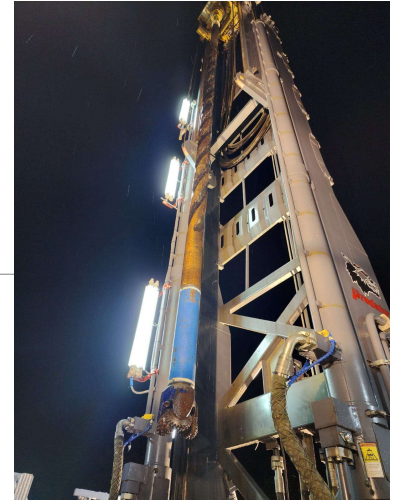
- HQ drilling, HW liner running
- NQ drilling, NW liner running

Slim – core hole

Best practices and lesson learned

Rotary Drilling

- Jet cellar system capacity is very critical to transfer out the return from cellar
- Hivis sweep to assist inadequate circulation for the hole cleaning
- Casing setting depth of the conductor, surface and production casing at the minimum acceptance level instead of pushing it deeper
- Casing running with the thread nubbins
- Keep a simple cementing with a top plug
- Be ready with the losses.
- With a limited capacity of the equipment and material, a plug cementing in a sequence and staging instead of one-time could be a best option.
- Cement plug design testing is crucial.
- Material selection of LCM pills, background lcm in the active system, natural LCM, and a bag of natural LCM are needed to prepare a head.



Slim – core hole

Best practices and lesson learned

Continuous Coring

- Need to plan a head of the coring equipment set up and pre-fabricate before the rig mobilize.
- Retrieve the core barrel immediately after it couldn't make progress to prevent the core head being worn out and spinning.
- Rod grease will be applied after the TLC. The suitable grease spec, and the practice to apply it are needed to assess.
- Corehead and BHA selection for TLC conditions needs to conduct.
- Inadequate RPM applied will lead to concave face wear rounded with under gauge of the corehead. Adequate parameters are important for BHA reliability and performance.
- Strongly recommend to run in back with all new BHA, and if possible, prevent any rerun BHA component.
- The rod connection is design for tension and not compression.

Slim – core hole

Best practices and lesson learned

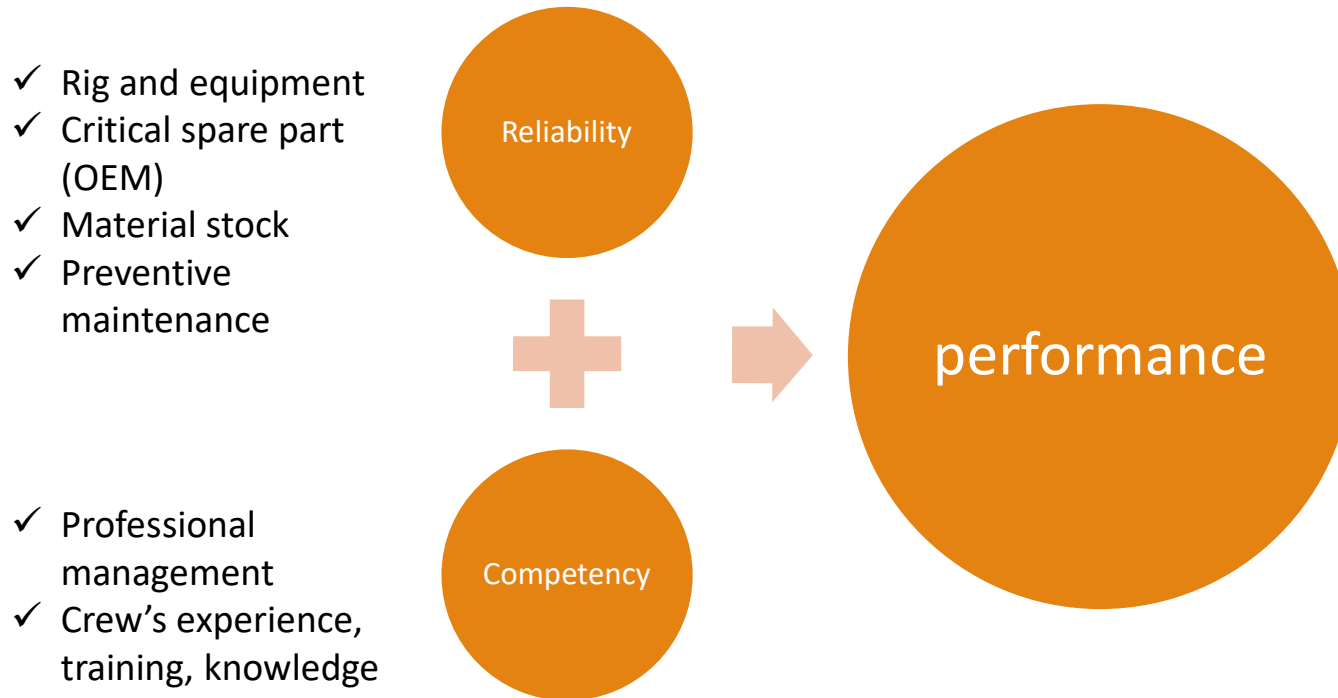
Continuous Coring

- Always prepare the cutter with the back up onsite and for all rod sizes. The cutter successfully cut the HRQ rod with no issue.
- The rod and top drive connections need to be inspected regularly with a visual and thread gauge. Make every connection if possible, considering the frequency of breakouts and making up the connection. Immediately laydown the pipe, which has a thread damage indication.
- Always install the safety clamp as the secondary, release the tread nubbins after ensuring the clamps are installed properly.
- The dies of the foot clamp and safety clamp condition deteriorate rapidly with the frequency of breakouts and making up the connection. Be prepared with extra back up.
- Need to have TDS Saver sub, both main and backup, for all drill-rod connection in GOOD condition
- The back up of a contingency a rod such as NRQ at least double of the TD length of the well
- Prepare a 1-7/8” dummy tool and 1-3/4” PTS
- The taper tap and spear of varying size need to be prepared for fishing. It isn't recommended to reconnect the rod because of the thread with big and flat design has low chance of being reconnected and high risk of thread damage.

Slim – core hole

Best practices and lesson learned

A professional, experience, and proven track record drilling contractor is the key success factor



Slim – core hole product knowledge

Boart Longyear - History



Edmund J. Longyear



It all started with a mining engineer in 1888, named Edmund J. Longyear
who drilled the first diamond drill core hole in northern Minnesota.

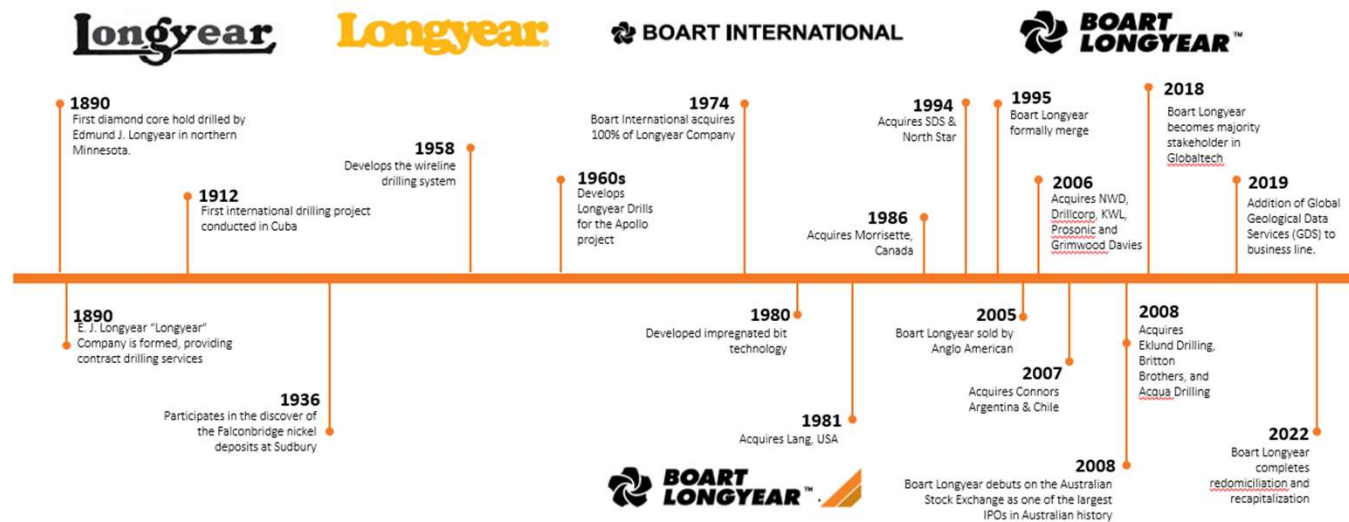
Slim – core hole product knowledge

Boart Longyear – History



Boart Longyear - History

BOART LONGYEAR AT A GLANCE



Boart Longyear - History

BOART LONGYEAR AT A GLANCE

Our company is truly multicultural, working in **over 30 countries!**



Boart Longyear - History

BOART LONGYEAR AT A GLANCE

Today, Boart Longyear is the world's leading provider of innovative drilling services, manufacturer of productivity-driven drilling equipment, and developer of orebody-data technology that is revolutionizing the future of mineral development. With its rich heritage from more than 130 years of operations, the company continues to build value for customers worldwide.



GEOLOGICAL DATA SERVICES is redefining the future of mineral exploration by providing customers with secure, accurate geological data to confidently make critical development decisions faster and more cost effective on site.



DRILLING SERVICES is the industry-leading global provider of drilling services to the mining sector as well as the world's largest, specialised, and diversified driller.



GLOBAL PRODUCTS offers the most advanced technology tooling and drill rigs, specifically engineered for long life and high performance while delivering world-class safety features.



DID YOU KNOW?

Boart is another name for the diamonds used in the drill bits.

Boart Longyear Products

LF70



LFTM70 DIAMOND CORE FLY DRILL

Mobility

The LFTM70 modular design consists of seven sections that are heliportable for reaching remote locations. This drill reduces downtime when mobilizing between sites and significantly decreases the environmental impact to a drill site.

Modular Design

With just seven components, it takes less than one hour to pull down the rig and the same amount of time to reassemble it.

Lightweight

Special attention was given to the overall weight and dimension of the drill's design. Each of its seven components weigh less than 585 kg and are easily transportable by helicopter. The LF70 is ideal for drilling in confined, remote locations.

Boart Longyear Products

LF70 TM70 TECHNICAL INFORMATION

Drilling Depth Guidelines				
Drill Rod / Core Barrel	Dry Hole		Fluid Filled	
	Hole Depth (meters)	Hole Depth (feet)	Hole Depth (meters)	Hole Depth (feet)
BRQ / BQ	907	2,976	1,039	3,407
BRQTK / BQTK	1,142	3,746	1,308	4,292
NRQ / NQ / NQ2	698	2,290	798	2,619
NRQ W-Wall TM	789	2,588	895	2,936
HRQ / HQ	473	1,553	542	1,778
HRQ W-Wall TM	596	1,955	674	2,210
PHD / PQ	313	1,026	357	1,171
PHD W-Wall TM	426	1,399	478	1,567

The figures in these table are estimates which have been calculated using the applicable pullback capacity of the drill and on an effective rock tensile strengths of 5 MPa. Actual drilling results may vary and will depend on in-hole tools, subsurface and other environmental conditions, drilling techniques and equipment used. Always verify manufacturers' rod depth ratings prior to use.

Prime Mover		
	Metric	U.S.
Standard Unit	Cummins QSB 4.5 L, 4 cylinder, turbo charged, after cooled diesel engine.	
Displacement	4.5 L	275 in3
Power (maximum) at 2,300 RPM	110 kW	148 hp
Emissions Certification	CARB Tier 3 - EU Stage III A	CARB Tier 3 - US EPA
Fly Weight	585 kg	1290 lbs

Boart Longyear Products

LF70

HQ Drill Head - Standard		
	Metric	U.S.
Standard HQ - Hollow Spindle		
Maximum Rod Diameter	95 mm	4 in
Rotation Motor	Rexroth hydraulic motor - variable/reversible	
Mechanical Transmission	Funk 4 speed	
Ratios	1st	6.63:1
	2nd	3.17:1
	3rd	1.72:1
	4th	1.00:1
Final Drive	Roller chain drive	
Ratio	2.58:1	
Head Opener	Pivot style — manual operation	
	Patented Nitro-Chuck™	
Hydraulic HQ Chuck	Hydraulically opened, nitrogen gas spring closed. Axial holding capacity of 133 447 N (30,000 lbf)	
Drill Head Lubrication	Force fed bearings, oil bath for roller chain - PTO driven	
Drill Head Lubrication Oil Filtration	25 Micron high pressure oil filter	
Fly Weight	390 kg	860 lb

Torque And RPM Ratings - HQ Head			
(Hydraulic motor at maximum/minimum displacement, prime mover at 2,200 RPM)			
	Speed (no load)	Torque (stall)	
	RPM	Nm	lbf
1st Gear	95 - 190	3,956 - 2,127	2,918 - 1,569
2nd Gear	200 - 400	1,891 - 1,017	1,395 - 750
3rd Gear	370 - 730	1,026 - 552	757 - 407
4th Gear	630 - 1,250	634 - 341	468 - 251
NOTE: Head output speed and torque are infinitely variable in each gear range as indicated. Actual rotation speed is affected by engine RPM and hydraulic motor displacement setting.			

PQ Drill Head - Optional		
	Metric	U.S.
Optional PQ - Hollow Spindle		
Maximum Rod Diameter	122 mm	5 in
Rotation Motor	Rexroth hydraulic motor - variable/reversible	
Mechanical Transmission	Funk 4 speed	
Ratios	1st	6.27:1
	2nd	3.12:1
	3rd	1.75:1
	4th	1.00:1
Final Drive	Straight cut gears	
Ratio	2:1	
Head Opener	Pivoting style: manual operation	
	Patented Nitro-Chuck™	
Hydraulic PQ Chuck	Hydraulically opened, nitrogen gas spring closed Axial holding capacity of 222 400 N (50,000 lbf)	
Drill Head Lubrication	Force fed bearings, oil bath for gears - PTO driven	
Drill Head Lubricating Oil Filtration	25 micron high pressure oil filter	
Fly Weight	580 kg	1,279 lb

Torque And RPM Ratings - PQ Head			
(Hydraulic motor at maximum/minimum displacement, prime mover at 2,200 RPM)			
	Speed (no load)	Torque (stall)	
	RPM	Nm	lbf
1st Gear	122 - 199	3,878 - 2,027	2,861 - 1,495
2nd Gear	246 - 400	1,929 - 1,008	1,423 - 743
3rd Gear	439 - 714	1,084 - 567	800 - 418
4th Gear	769 - 1,250	658 - 344	485 - 253
NOTE: Head output speed and torque are infinitely variable in each gear range as indicated. Actual rotation speed is affected by engine RPM and hydraulic motor displacement setting.			

Hydraulic System		
Primary Pump	Axial piston, variable displacement load sensing, pressure compensated with low pressure standby.	
Max Flow	163 L/m	43 gpm
Maximum Pressure (factory setting)	241 bar	3,500 psi
Secondary Pump	Axial piston, variable displacement, pressure compensated	
Max Flow	42 L/m	11 gpm

Boart Longyear Products

LF70

Hydraulic System (continued)		
Maximum Pressure (factory setting)	138 bar	2,000 psi
Auxiliary Pump	Axial piston, variable displacement pressure compensated.	
Max Flow	42 L/m	11 gpm
Maximum Pressure (factory setting)	138 bar	2,000 psi
Hydraulic Tank Capacity	114 L	30 Gal

Drill Mast And Feed System		
	Metric	U.S.
Feed Stroke	1.8 m	6 ft
Max Pull Capacity	78.6 kN	17,671 lbf
Max Thrust Capacity	51.9 kN	11,658 lbf
Rod Pull	3 or 6 m	10 or 20 ft
Drilling Angle	45° off horizontal to 90° vertical down	
Mast Dump (Crowd)	N/A	

Draw Works		
	Metric	U.S.
Main Line Hoist	Main Line Hoist Single speed motor	
Hook Load (single part line)		
Bare Drum	53.4 kN	12,000 lb
Hoisting Speed (single part line)		
Bare Drum	1 m/sec	3 ft/sec
Main Hoist Cable	15 mm	0.59 in
Minimum Breaking Strength	226 kN	50,800 lb
Note: Do not use multiple part lines with the main line hoist, use single part line only.		
Foot Clamp Capacity	HWT	
Wireline Hoist		
Level Wind	N/A	
Line Pull		
Bare Drum	9.7 kN	2,190 lb
Full Drum	2.2 kN	502 lb
Line Speed		
Bare Drum	100 m/min	337 ft/min
Full Drum	433 m/min	1,470 ft/min
Drum Capacity 4.8 mm (3/16") swaged	1,890 m	6,200 ft
Minimum Breaking Strength	22.2 kN	5,000 lb
Note: Wireline cable length to be specified at time of order		

Additional Information		
	Metric	U.S.
Fuel Tank Capacity	57 L	15 Gal

Weight - Fly Drill = 2,950 kg (6,500 lbs) HQ Version / 3,740 kg (8,245 lbs) PQ Version		
Consisting of:		
Cummins 4.5 L QSB, Tier 3, 4 cylinder, water cooled, turbo charged, diesel engine		
Hydraulic Module		
Draw Works Grp. c/w 12,000 lb Main Line Hoist, Wireline Hoist		
3 Piece Lattice Mast Assembly		
HQ Rotation Unit Grp. c/w Nitro-Chuck™		
Base Frame		
Fuel Tank (57 L/15 US gal)		
Battery - 12V		
Stabilizer Legs (4)		

Options		
PQ rotation unit group		
Decals available in multiple languages		
Mast raising cylinders		
Wireline cable		
Rod rack - WARNING: Do not operate this drill with rods racked in wind velocities in excess of 85 km/h		
Mud tank		
Towing package - NOTE: Not highway rated		
Fluid circulation pumps (diesel supply and pressure)		
Mud mixer		

Boart Longyear Products

LF90



LFTM90D SURFACE CORING DRILL

Mobility

Engineered for mobility, the LFTM90D surface coring drill features a telescoping mast with dump capability. The telescoping features afford a compact size during transport while the dump capability lowers the working height of the machine to enable mounting on a crawler, truck or skid.

Reliability

A simple hydraulic and structural design makes this drill easy to operate and maintain.

Safety

Designed with safety in mind, the LF90D surface coring drill includes standard safety features such as an interlocked rotation barrier that slows rotation when barrier is opened and a side shifting head which reduces working height when handling inner tubes.

Boart Longyear Products

LF90

Drilling Depth Guidelines		
	Fluid Filled	
Drill Rod / Core Barrel	Hole Depth (meters)	Hole Depth (feet)
BRQ / BQ	1 385	4,543
NRQ / NQ	1 064	3,492
NRQ V-WALL™	1 193	3,915
HRQ / HQ	722	2,370
HRQ V-WALL™	898	2,947
PHD / PQ	476	1,562
PHD V-WALL™	637	2,089

The figures in these table are estimates which have been calculated using the applicable pullback capacity of the drill and on an effective rock tensile strengths of 5 MPa. Actual drilling results may vary and will depend on in-hole tools, subsurface and other environmental conditions, drilling techniques and equipment used. Always verify manufacturers' rod depth ratings prior to use.

Prime Mover		
	Metric	U.S.
Standard Unit	Cummins QSB 6.7 L, liquid cooled, turbo charged, after cooled, diesel engine	
Displacement	6.7 L	409 in³
Power (maximum) at 2200 RPM	153 kW	220 hp
Emissions Certification	Stage III	Tier 3

Torque and RPM Ratings			
(Hydraulic motor at maximum/minimum displacement, prime mover at 2,200 RPM)			
	Speed (no load)	Torque (stall)	
	RPM	Nm	lbf ft
1st Gear	122 - 199	5 322 - 3 254	3,925 - 2,400
2nd Gear	246 - 400	2 648 - 1 620	1,963 - 1,195
3rd Gear	439 - 714	1 486 - 908	1,096 - 670
4th Gear	769 - 1,250	849 - 519	626 - 383
NOTE: Head output speed and torque are infinitely variable in each gear range as indicated. Actual rotation speed is affected by RPM and hydraulic motor displacement setting.			

Hydraulic System		
	Metric	U.S.
Primary Pump	Axial piston, variable displacement load sensing, pressure compensated with low pressure standby.	
Max Flow	165 L/m	43.5 gpm
Max Pressure (factory setting)	31 MPa	4,500 psi
Secondary Pump	Axial piston, variable displacement load sensing, pressure compensated with low pressure standby.	
Max Flow	64 L/m	17 gpm
Max Pressure (factory setting)	21 MPa	3,000 psi
Auxiliary Pump	Axial piston, variable displacement, pressure compensated	
Max Flow	42 L/m	11 gpm
Max Pressure (factory setting)	14 MPa	2,000 psi
Hydraulic Tank Capacity	231 L	61 Gal

Drill Head		
Rotation Motor	Rexroth hydraulic motor - variable/reversible	
Mechanical Transmission	Funk 4 speed	
Ratios	1st	6.27:1
	2nd	3.12:1
	3rd	1.75:1
	4th	1.00:1
Final Drive	Straight cut gears	
Ratio	2:1	
Head Opener	Side shift style - hydraulically actuated	
	Patented Nitro Chuck™	
Hydraulic PQ Chuck	Hydraulically opened, nitrogen gas spring closed	
	Axial holding capacity of 222 400 N (50,000 lbf)	
Drill Head	Force fed bearings, oil bath for gears	
Drill Head Lubricating Oil Filtration	25 micron suction oil filter - independent constant flow	

Drill Mast and Feed System		
	Metric	U.S.
Feed Stroke	3.35 m	11 ft
Max Pull Capacity	167.7 kN	37,700 lbf
Max Thrust Capacity	88.4 kN	19,880 lbf
Rod Pull	3 or 6 m	10 or 20 ft
Drilling Angle	45° off horizontal to 90° vertical down	
Mast Dump (Crowd)	2.34 m	7.7 ft
Mast Telescope	3.35 m	11 ft

Boart Longyear Products

LF90

Draw Works		
	Metric	U.S.
Main Line Hoist	Single speed motor	
Hook Load (single part line)		
Bare Drum	71.2 kN	16,000 lbf
Hoisting Speed (single part line)		
Bare Drum	.35 m/min	1.15 ft/s
Main Hoist Cable	15 mm	0.59 in
Minimum Breaking Strength	226.0 kN	50,800 lb
Note: Do not use multiple part lines with the main line hoist, use single part line ONLY.		
Foot Clamp Capacity	HWT	
Wireline Hoist		
Level Wind	N/A	
Level Pull		
Bare Drum	12 kN	2,700 lb
Full Drum	2.2 kN	502 lb
Line Speed		
Bare Drum	123 m/min	405 ft/min
Full Drum	321 m/min	1054 ft/min
Drum Capacity (4.8 mm/316" swaged)	1 890 m	6,200 ft
Minimum Breaking Strength	2 268 kg	5,000 lb
Note: Wireline cable length to be specified at time of order		

Additional Information		
	Metric	U.S.
Fuel Tank Capacity	243 L	64 Gal

* Dimensions and weights may vary depending on options and should be checked before crating or lifting.

Weight = 8,392 KG	
Consisting of:	
Power Unit Group Cummins QSB 6.7 L, Tier 3, 6 cylinder	
Hydraulic Module	
Draw Works Grp. c/w 16,000 lb Main Line Hoist with Cable, Wireline Hoist less Cable	
Hydraulic Mast Raising	
Hydraulic Mast Dump	
Telescopic Mast Assembly	
Rotation Unit Grp. c/w PQ Nitro Chuck™/Head Guard	
Base Frame	
Fuel Tank (265 L/70 US gal)	
Battery - 12V	
Hydraulic Foot Clamp HWT Capacity	

Options	
Truck	NOTE: The LF90D requires an additional sub frame for mounting onto a truck.
Catwalk, Railing and Access Stairs	
Rod Rack Platform	
Hydraulic Leveling Jacks/Outriggers	
Truck Mounting Sub-frame	
Hot Weather Hydraulic Cooling Group	WARNING: Do not operate this drill with rods racked in wind velocities in excess of 85 km/h.
CE Certification	
Decals available in multiple languages	
Fluid circulation pumps	
Mud Mixer	

Drill Transport and Position -Basic Drills	
Wet weight = 18600 lbs (8437 kg)	
Ground pressure: 0.6 kg/cm ²	
Max speed: 2.4 kph	

Boart Longyear Products

LF230 Track Mounted



LF™ 230 SURFACE CORING DRILL

Power

The LF™230 deep hole drill provides the features and the power required to get just about any job done, easily and efficiently. A 40,000 lb hoist allows for rig depth capacity beyond its class. The PQ™ Nitro-Chuck™ is field tested and designed to offer performance and reliability to conquer even the hardest projects.

Mobility

The LF230 is contained to a small footprint, optimized for easy transport. Engineered for mobility, the LF230 surface coring drill features a telescoping mast with dump capability. The telescoping features afford a compact size during transport while the dump capability lowers the working height of the machine to enable mounting on a truck or scow.

Reliability

The open-faced mast features a common center line through the chuck and feed cylinders, decreasing wear on components. Simple hydraulics and structural design makes this drill easy to operate and maintain.

Boart Longyear Products

LF230 Track Mounted

Drilling Depth Guidelines		
Fluid Filled	Metric	U.S.
Drill Rod / Core Barrel	Hole Depth Meters	Hole Depth Feet
BRQ™ / BQ™*	3462	11358
NRQ™	2661	8730
NRQ™ V-Wall™	2983	9788
HRQ™ / HQ™	1806	5925
HRQ™ V-Wall™	2246	7368
PHD /PQ™	1190	3904
PHD V-Wall™	1592	5223
*The figures in these tables have been calculated using the pulling capacity of the mainline hoist with a fluid filled hole and an effective rock tensile strength of 5MPa. Actual drilling results may vary and will be affected by in-hole tools, fluid level in the hole, subsurface and other environmental conditions, drilling techniques, and equipment used.		
* BQ capacity is shown for comparison purposes only. It is not recommended drilling practice to drill over 2000 m BQ depth. Always verify manufacturers drill rod depth ratings prior to use.		
Prime Mover		
	Metric	U.S.
Standard Unit	Cummins QSC 8.3 L, liquid cooled, turbo charged, charge air cooled engine.	
Displacement	8.3 L	506 in³
Power (maximum) at 2,200 RPM	205 kW	275 hp
Emissions	Stage™ III	tier 3
Torque and RPM Ratings		
	Metric	U.S.
(Hydraulic motor at maximum/minimum displacement, prime mover at 2,200 RPM)		
	Speed (no load)	Torque (stall)
	RPM	Nm ftlbs
1st gear	144 - 199	5322 - 3826 3925 - 2822
2nd gear	288 - 400	2648 - 1898 1953 - 1400
3rd gear	514 - 714	1486 - 1068 1096 - 788
4th gear	900 - 1250	849 - 610 626 - 450
Note: Head output speed and torque are infinitely variable in each gear range as indicated.		
Actual rotation speed is affected by engine RPM and hydraulic motor displacement setting.		

Hydraulic System		
	Metric	U.S.
Primary Pump	Axial piston, variable displacement, load sensing, pressure compensated with low pressure standby.	
Max flow	318 L/m	84 gpm
Maximum Pressure (factory setting)	31 MPa	4500 psi
Secondary Pump	Axial piston, variable displacement, load sensing, pressure compensated with low pressure standby.	
Max flow	72 L/m	19 gpm
Maximum pressure (factory setting)	21 MPa	3000 psi
Auxiliary Pump	Axial piston, variable displacement, pressure compensated.	
Max flow	42 L/min	11 gpm
Maximum pressure (factory setting)	14 MPa	2000 psi
Hydraulic Tank Capacity	435 L	115 gal
Drill Head		
	Metric	U.S.
Rotation Motor	Resroth hydraulic motor - variable / reversible	
Ratios	1st	6.27:1
	2nd	3.12:1
	3rd	1.75:1
	4th	1.00:1
Final Drive	Straight cut gears	
Ratio	2:1	
Head Opener	Pivoting style - hydraulically actuated	
Hydraulic PQ™ Chuck	Patented Nitro-Chuck™ Hydraulically opened, nitrogen gas spring closed Axial holding capacity of 222 400 N (50,000 lbf)	
Drill Head Lubrication	Force fed bearings, oil bath for gears, external sump	
Drill Head Lubricating Oil Filtration	25 micron suction oil filter	
Drill Mast and Feed System		
	Metric	U.S.
Feed stroke	3.35 m	11 ft
Max Pull Capacity	223.6 kN	50,265 lbf
Max Thrust Capacity	117.9 kN	26,500 lbf
Rod Pull	6 or 9 m	20 or 30 ft
Drilling Angle	45° off horizontal to 90° vertical down	
Mast Dump (crowd)	2.74 m	9 ft
Mast Telescope	2.87 m	9.42 ft

U.S. Patent No. 6,719,993; CA Patent No. 2,376,972; AU Patent No. 2002010146; Patent pending.
All other party trademarks are the property of their respective owners.

Boart Longyear Products

LF230 Track Mounted

Draw Works		
	Metric	U.S.
Main line winch	Two speed motor	
Hook load (single part line)		
Bare Drum	177.9 kN	40000 lb
Hoisting speed (single part line)		
Bare Drum	1.1 m/s & .67 m/s	3.6 ft/s & 2.2 ft/s
Main line winch cable dia.	22 mm	7/8 in
Minimum breaking strength	508.8 kN	114400 lbf
Note: Do not use multiple part lines with the main line hoist, use single part line ONLY.		
Foot clamp capacity	HWT - 178kN (Axial)	
Wireline Hoist		
Level wind	Chain driven - dual ratio	
Line Pull		
Bare Drum	17.2 kN	3880 lbs
Full Drum	9.5 kN	2130 lbf
Line Speed		
Bare Drum	160 m/min	525 ft/min
Full Drum	284 m/min	932 ft/min
Drum Capacity	3300 m	10827 ft
Note: Wireline cable length to be specified at time of order		

Additional Information		
Fuel Tank Capacity	Metric	U.S.
Standard	564 L	149 gal

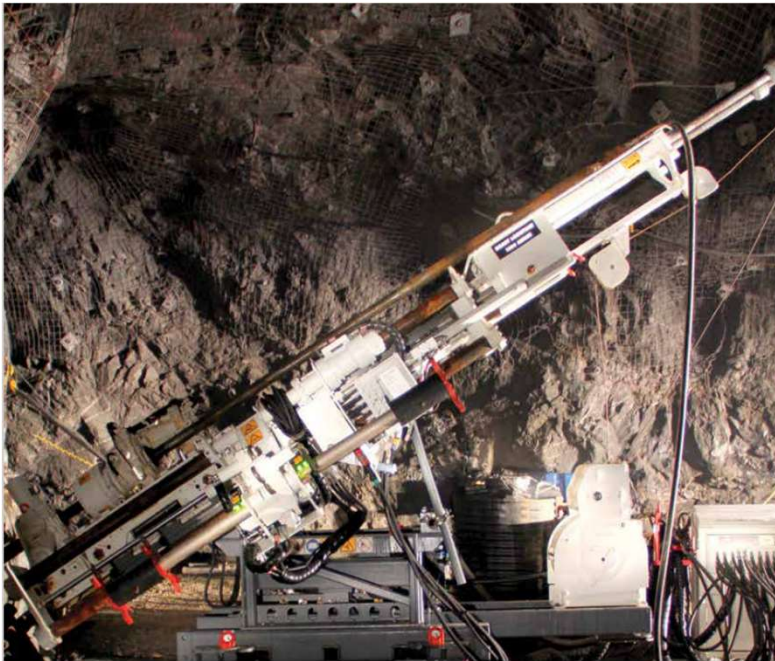
DIMENSIONS AND WEIGHTS*

Wet Weight **=13807 kg (32160 lb)	
Consisting of:	
Power unit group Cummins QSC 8.3 L, tier 3, 6 cylinder	
Hydraulic Module	
Draw Works Grp. c/w 40,000 lb main line hoist with cable, wireline hoist less cable	
Hydraulic mast raising	
Hydraulic mast dump	
Telescopic mast assembly	
Rotation unit grp. c/w PQ™ NitroChuck™	
Base frame	
Fuel tank (485 L/128 gal)	
Battery	
Hydraulic leveling jacks / outriggers (optional)	
Foot clamp	
Mast access ladders	
Hydraulic water pump	
** For high mobility, the mast mounted platform is kept lightweight. An optional rod rack is available for vertical holes only, otherwise rods are stacked on the ground.	
Options	
Truck mounted - International 5600	
Catwalk, railing and access stairs	
Rod rack platform (vertical holes only)***	
Hydraulic cooler (hot climate)	
Mast rest (truck mount version)	
Decals available in multiple languages	
Fluid circulation pumps (diesel supply and pressure)	
Mud mixer	
*** Limited to 22000 lb of gross weight stacked at 5° off vertical mast, i.e. 140 x 30 ft stands NQ™	
Warning: do not operate this drill with rods racked in wind velocities in excess of 85 km/h. this unit must not be operated without a truck or drilling skid base installed for stability.	

* Dimensions and weights may vary depending on options and should be checked before crating or lifting.

Boart Longyear Products

LM90



LM™ 90 UNDERGROUND CORING RIG

The LM™90 is powered by a 90 kW electric motor. It is modularly designed like the other LM rigs allowing for easy tailoring and upgrading as specific needs and conditions change.

With the assistance of a positioner and a turntable, the drill is capable of drilling holes at all angles from vertically up to vertically down.

MODULAR DESIGN

The LM90 was designed as a modular drill rig with a number of options for site-specific customization. This drill is easy to use and upgrade to meet the needs of future drilling projects.

RELIABLE

The LM90 features a load sensing hydraulic system, proportional controls, a fail-safe rod clamp, built in troubleshooting and a spooling device. These features make the LM90 a highly reliable drill when drilling in harsh conditions.

SAFE

An optional patented rod handler can be added to the LM90, which greatly increases driller safety. The rod handler makes loading rods easier and ensures they remain away from moving parts during rod tripping.

Boart Longyear Products

LM90

Specification	Metric	U.S.
Vertical Depth Capacity (1300 FF)	1,400 m (NRQ)*	4,593 ft (NRQ)*
Drill Rod Size	AQ to HQ	
Feed Force	63.6 kN	14,291 lb
Pull Force	127.7 kN	28,708 lb
Feed Length	1.83 m	6 ft
Head Torque	257 N-m @ 1,200 RPM 845 N-m @ 500 RPM	190 ft-lb @ 1,200 RPM 623 ft-lb @ 500 RPM
Electric Motor	90 kW	120 hp

For complete technical data and customization options, visit www.BoartLongyear.com/LM90

Boart Longyear – Coring Rods Product



Boart Longyear – Coring Rods Product

RQ™ W-WALL™ TECHNICAL INFORMATION

NRQ W-Wall Performance Rating		
	Metric	U.S.
Rated Drilling Depth by Joint Strength	3 300 m	10,800 ft
Rated Pullback	330 kN	75,000 lbf
Rated Torque (Operating or Make-Up)	2 400 Nm	1,750 ft-lb
Recommended Minimum Make-Up Torque for Deep Holes over 2000m	1 400 Nm	1,000 ft-lb
Recommended Minimum Make-Up Torque for Deep Holes over 1000m	1 000 Nm	750 ft-lb
Recommended Minimum Make-Up Torque	600 Nm	452 ft-lb
API Theoretical Burst Pressure at Box Shoulder (per API bulletin 5C3)	49 700 kPa	7,215 psi
API Theoretical Burst Pressure at Midbody (per API bulletin 5C3)	63 387 kPa	9,193 psi
API Theoretical Collapse Pressure at Midbody (per API bulletin 5C3)	52 421 kPa	7,603 psi

NRQ W-Wall Weights & Measures		
	Metric	U.S.
Rod Midbody Outer Diameter	69.90 mm	2.75 in
Rod Midbody Inner Diameter (maximum)	62.00 mm	2.44 in
Rod Joint Inner Diameter	60.30 mm	2.38 in
Rod Stiffness (Moment of Inertia)	11 408 mm ⁴	17.7 in ⁴
Rod Weight	20.4 kg/3 m	45 lb/10 ft
Avg. Content Weight (Water) per Unit Length	2.97 L/m	0.24 gal/ft

Boart Longyear – Coring Rods Product

HRQ W-Wall Performance Rating		
	Metric	U.S.
Rated Drilling Depth by Joint Strength	3 050 m	10,000 ft
Rated Pullback	510 kN	115,000 lbf
Rated Torque (Operating or Make-Up)	3 500 Nm	2,600 ft-lb
Recommended Minimum Make-Up Torque for Deep Holes over 2000 m	2 750 Nm	2,000 ft-lb
Recommended Minimum Make-Up Torque for Deep Holes over 1000 m	2 000 Nm	1,500 ft-lb
Recommended Minimum Make-Up Torque	1 000 Nm	750 ft-lb
API Theoretical Burst Pressure at Box Shoulder (per API bulletin 5C3)	44 500 kPa	6,458 psi
API Theoretical Burst Pressure at Midbody (per API bulletin 5C3)	50 200 kPa	7,280 psi
API Theoretical Collapse Pressure at Midbody (per API bulletin 5C3)	29 00 kPa	4,206 psi

HRQ W-Wall Weights & Measures		
	Metric	U.S.
Rod Midbody Outer Diameter	88.90 mm	3.50 in
Rod Midbody Inner Diameter	81.00 mm	3.19 in
Rod Joint Inner Diameter	77.80 mm	3.06 in
Rod Stiffness (Moment of Inertia)	2 4592 mm ⁴	38.1 in ⁴
Rod Weight	27.3 kg/3 m	60.0 lb/10 ft
Avg. Content Weight (Water) per Unit Length	5.14 L/m	0.40 gal/ft

Boart Longyear – Casing Product

HW

METRIC	ITEM	DESCRIPTION	OD (mm)	ID (mm)	WEIGHT (kg/3 m)	THREAD PITCH (mm)	PIN LENGTH (mm)	CONTENT (l/100 m)
	66551	HW 3.0 m CASING						
	66549	HW 1.5 m CASING	114.30	101.60	52.20	6.4	76.20	810.80
	66548	HW 1.0 m CASING						

IMPERIAL	ITEM	DESCRIPTION	OD (in)	ID (in)	WEIGHT (lb/10 ft)	THREAD PITCH (in)	PIN LENGTH (in)	CONTENT (g/100 ft)
	26370	HW 10' CASING						
	26371	HW 5' CASING	4.50	4.00	117.00	4.0	3.00	65.30
	26373	HW 2' CASING						
	26374	HW 1' CASING						

HW BUNDLE SPECIFICATIONS

3.0 m/10 ft CASING BUNDLE (7 PIECES)

Dimensions (L x W x H).....3.2 x 0.4 x 0.3 m (10.3 x 1.2 x 1.1 ft)
 Volume.....0.4 m³ (14.1 ft³)
 Gross Weight.....373 kg (823 lb)

1.5 m/5 ft CASING BUNDLE (7 PIECES)

Dimensions (L x W x H).....1.6 x 0.4 x 0.3 m (5.3 x 1.2 x 1.1 ft)
 Volume.....0.2 m³ (7.1 ft³)
 Gross Weight.....194.1 kg (428 lb)

CONTAINER SHIPMENTS:

20 ft container load of 3.0 m/10 ft rods holds 24 bundles (168 rods)
 40 ft container load of 3.0 m/10 ft rods holds 54 bundles (378 rods)

PW

METRIC	ITEM	DESCRIPTION	OD (mm)	ID (mm)	WEIGHT (kg/3 m)	THREAD PITCH (mm)	PIN LENGTH (mm)	CONTENT (l/100 m)
	66731	PW 3.0 m CASING						
	66730	PW 1.5 m CASING	139.70	127.00	64.30	6.4	82.55	1266.60
	66729	PW 1.0 m CASING						

IMPERIAL	ITEM	DESCRIPTION	OD (in)	ID (in)	WEIGHT (lb/10 ft)	THREAD PITCH (in)	PIN LENGTH (in)	CONTENT (g/100 ft)
	26629	PW 10' CASING						
	26630	PW 5' CASING	5.50	5.00	144.00	4.0	3.25	102.00
	26631	PW 2' CASING						
	26632	PW 1' CASING						

PW BUNDLE SPECIFICATIONS

3.0 m/10 ft CASING BUNDLE (7 PIECES)

Dimensions (L x W x H).....3.2 x 0.5 x 0.4 m (10.3 x 1.5 x 1.3 ft)
 Volume.....0.6 m³ (21.2 ft³)
 Gross Weight.....463 kg (1022 lb)

1.5 m/5 ft CASING BUNDLE (7 PIECES)

Dimensions (L x W x H).....1.6 x 0.5 x 0.4 m (5.3 x 1.5 x 1.3 ft)
 Volume.....0.3 m³ (10.6 ft³)
 Gross Weight.....254 kg (560 lb)

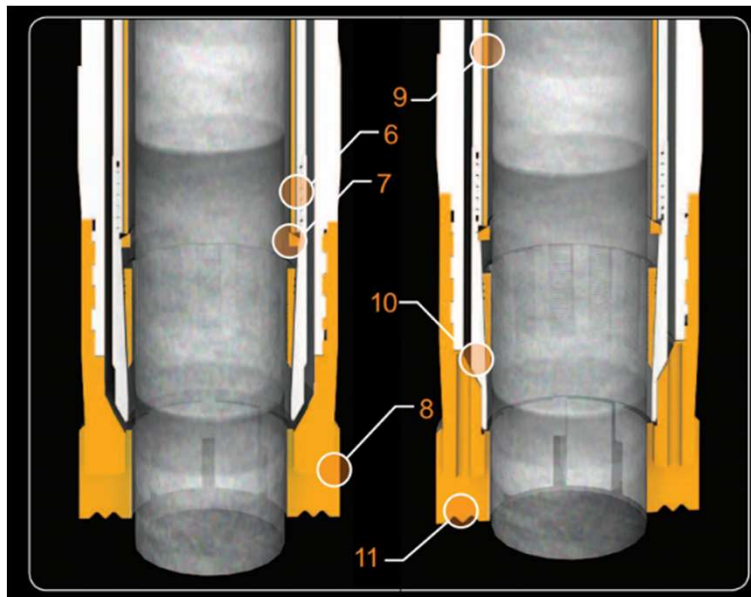
CONTAINER SHIPMENTS:

20 ft container load of 3.0 m/10 ft rods holds 20 bundles (140 rods)
 40 ft container load of 3.0 m/10 ft rods holds 44 bundles (308 rods)

Boart Longyear – Bit Product



Boart Longyear – Bit Product



Q³ SYSTEM OPTION

The Q³ system is designed to improve core recovery in soft or highly broken ground conditions.

STOP RING (6)

Retains the inner-tube and the liner.

INNER TUBE LINER (7)

The inner-tube liner is used when collecting highly fractured or soft core samples.

Q³ CORE DRILLING BIT (8)

Core bits made for a 3 tube core barrel system are designed to provide a smaller-diameter core sample which allows the core to feed into the extra tube.

Q^{TT} SYSTEM OPTION

The Q^{TT} system is similar to the Q³ system but includes a specialized core lifter case and bit. These components enable change in the water flow by creating a seal which routes water flow away from the inner-tube – further improving core recovery.

INNER TUBE LINER (9)

The inner-tube liner is used to improve core collection of highly fractured and soft core samples.

CORE LIFTER CASE (10)

A specially designed piloted core lifter case that improves core recovery and reduces core washing by directing fluid flow away from the ID of the core bit.

Q^{TT} CORE DRILLING BIT (11)

Core bits made for a 3-tube core barrel system are designed to accommodate a pilot core lifter case. This style of core bit requires face discharge ports or deep ID waterways.

Boart Longyear – Bit Product

PRODUCT RANGE

IMPREGNATED DIAMOND BITS
The impregnated diamond bit is used for cutting holes in medium to ultra hard formations. A variety of variations are available to maximize drill life and penetration.

The crown is made up of a metal powder and diamond matrix that wears away during the drilling process and exposes new layers of diamonds, renewing the cutting points.

SURFACE SET CORING BITS
Surface set bits are typically used in softer formations. They utilize a single layer of natural diamonds set in a hard matrix and can be manufactured in many crown profiles to fit the different types of formations.

TSD BITS
The TSD bit is used for fast penetration in soft/medium rock formations. TSD's are a cost effective alternative for large stone surface set bits.

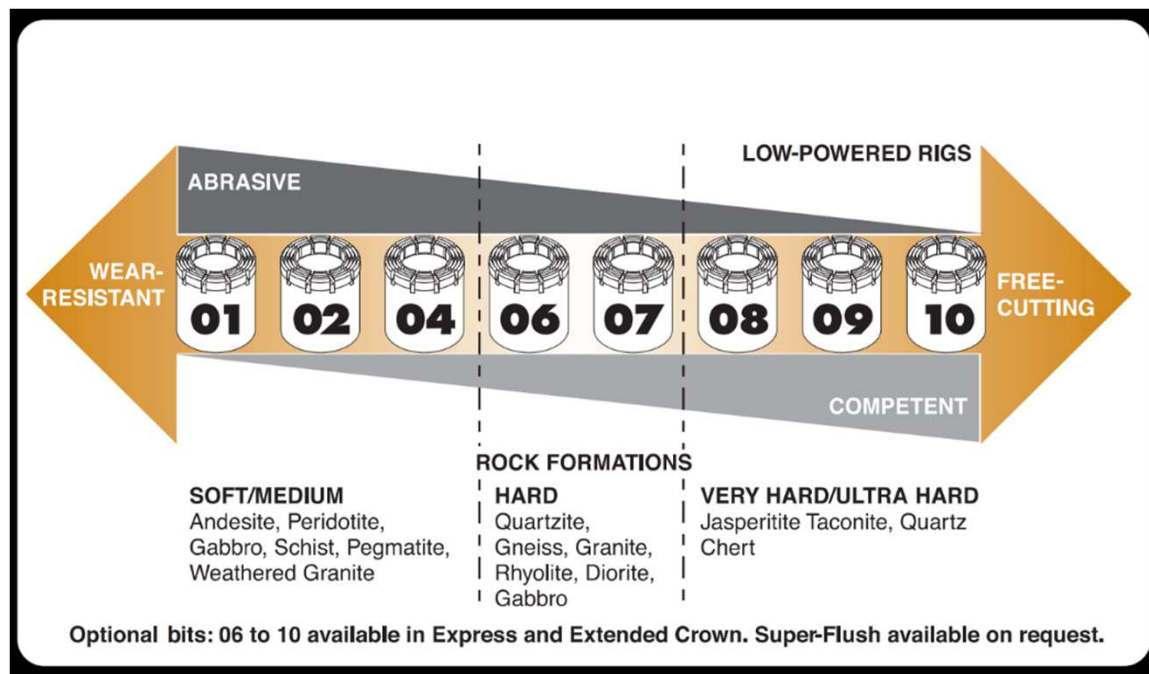
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PCD BITS
PCD bits are made for cutting softer formations at high penetration rates. They typically have round PCD cutters that can be replaced, extending the life of the bit.

CARBIDE CHIP
The carbide chip bit is a general purpose bit for geo-technical applications used primarily in soft formations. This is a cost effective alternate to carbide tooth bits.

DIRECTIONAL BITS
Directional bits are used with down hole motors to steer the drill string in different directions.

Boart Longyear – Bit Product



Boart Longyear – Down Hole Gear Product

SYSTEM OVERVIEW



OVERSHOT ASSEMBLY (1)

The overshot is dropped, or pumped, into the drill string to retrieve the inner tube assembly via wireline cable and hoist.

LOCKING COUPLING (2)

The 'Locking Coupling' threads to the drill rod string and provides a hardened mating surface against which the core barrel inner tube assembly latches ride while drilling.

Additionally, Locking couplings provide directional control for the core barrel assembly through wear resistant stabilizing pads which abrade against the drill hole wall.

ADAPTER COUPLING (3)

The adapter coupling mates between the locking coupling and core barrel outer tube, providing the pocket into which the head assembly latches deploy.

HEAD ASSEMBLY (4)

The head assembly provides: latching and pivoting spearpoint mechanisms to allow insertion and retrieval of the inner tube assembly; a bearing assembly to allow the inner tube to remain stationary and avoid sample damage while drilling; fluid pressure operating indications and fluid control valves.

All head assemblies incorporate a shut off valve assembly which provides a fluid pressure signal to the drill operator when the valve members are compressed, indicating a full or blocked inner tube.

OUTER TUBE (5)

The outer tube houses the inner tube assembly and connects to the diamond products cutting the hole. The increased wall thickness of the outer tube provides additional stiffness for directional control and a tighter hole annulus for increased fluid velocity and rapid cuttings evacuation for bit performance.

Multiple outer tubes can be assembled to extend the possible core sample length.

INNER TUBE (6)

The inner tube captures the core sample as drilling progresses. Multiple inner tubes can be assembled with couplers or extensions to accept longer core samples.

INNER TUBE STABILIZER (7)

Seated in the reaming shell, or in mated outer tube extensions, the replaceable and reversible inner tube Stabilizer provides centralizing for improved sample recovery, and a bearing between the stationary inner tube and the rotating outer tube.

CORE LIFTER (8)

The core lifter is a hardened steel, split collar with a tapered body that mates to a tapered socket in the core lifter case.

In a core breaking operation, the drill string is lifted off bottom and the core sample begins to slide out of the inner tube. Grip features on the inner surface of the core lifter catch the moving core sample and pull the core lifter towards the smaller end of the tapered socket in the core lifter case. The core lifter is constricted against the core sample and retains it after it has broken, allowing retrieval to surface.

CORE LIFTER CASE (8)

The core lifter case mates to the inner tube and houses the core lifter in a tapered socket which controls movement of the core lifter.

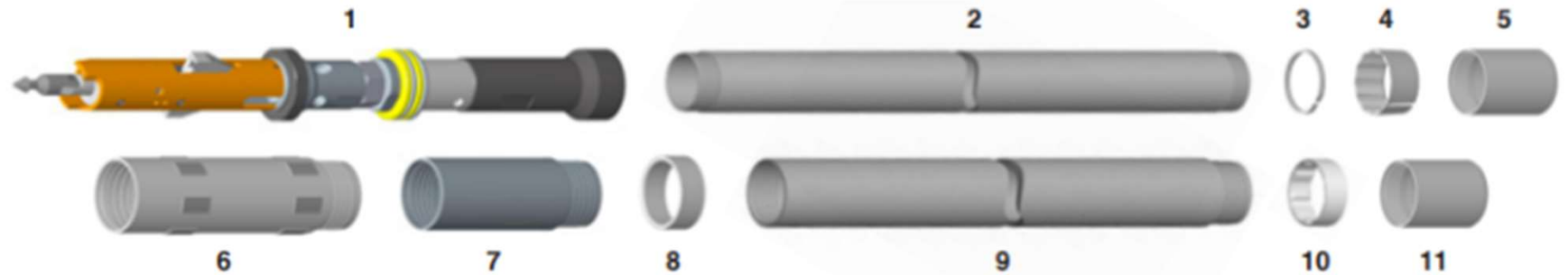
As the drill string is lifted during a core breaking operation, the core lifter case bottoms out on the inside of the drill bit transferring the pullback load from the drill string to the core lifter until the core sample breaks.

STOP RING (9)

The Stop Ring is hardened steel snap ring designed to seat into a mating groove, and retain the core lifter in the core lifter case.

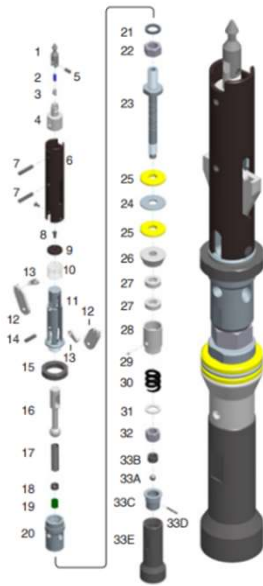


Boart Longyear – Down Hole Gear Product



Boart Longyear – Back End / Head Assembly Product

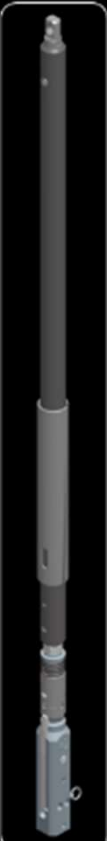
ID#	PART #	DESCRIPTION	QTY	KG	LB
	5001444	PQ HEAD ASSEMBLY		5.54	11.77
1	15141	SPRING, COMPRESSION	1	0.19	0.42
2	42914	PLUNGER, DETENT	1	0.10	0.22
3	3540330	BASE, HQ SPEARHEAD	1	0.01	0.02
4	42905	PIN, SPIROL 7/16" X 1"	1	0.77	1.69
5	5001332	CASE, HQ LATCH RETRACTING	1	0.01	0.02
6	24524	PIN, SPRING 1/2" X 2-3/4"	1	2.88	6.34
7	3541932	C/SCR, HH 3/8-16 3/4" G8/NYLOK	2	0.08	0.18
8	104953	WASHER, HQ	1	0.05	0.11
9	3543619	SPRING, HQ/PQ COMPRESSION	1	0.18	0.40
10	5001342	UPPER LATCH BODY	1	0.04	0.09
11	3543796	LATCH, PQ	1	2.45	5.39
12	306216	LINK, HQ	2	0.76	1.67
13	24305	PIN, SPRING 1/2" X 2"	2	0.12	0.26
14	3543797	LANDING SHOULDER, PQ	1	0.03	0.07
15	3548544	PISTON, HQ-PQ Q/P VALVE	1	0.74	1.63
16	3548718	PIN, SPIROL 1/4 X 2 3/4 LG	1	0.12	0.26
17	306131	BUSHING, FLUID RETENTION	1	0.02	0.04
18	104966	SPRING, FLUID RETENTION LIGHT	1	0.03	0.07
19	3548072	BODY, HQ/PQ Q/D LOWER LATCH	1	0.03	0.07
20	3545206	WASHER, NORD-LOCK 1-1/2"	1	0.00	0.00
21	37475	NUT, HEX 1-1/2 UNC G8	1	0.00	0.00
22	3548073	SPINDLE, HQ/PQ Q/D HOLLOW	1	0.41	0.90
23	3540308	VALVE, PQ HARD SHUT-OFF	1	2.10	4.62
24	3540300	WASHER, SHUT OFF VALVE	2	0.20	0.44
25	3540299	WASHER, TAPERED SHUT-OFF PQ	2	0.00	0.00
26	24528	BEARING, BALL THRUST	1	0.00	0.00
27	3540794	SPINDLE BUSHING, HQ/PQ Q/D	3	0.69	1.52
28	17447	FITTING, HYD GREASE	1	1.66	3.65
29	3540755	SPRING, COMPRESSION	1	0.01	0.02
30	3544324	WASHER, FLAT 1-1/4" NARROW	1	0.43	0.95
31	103062	L/NUT, 1-1/4 UNC NYLOK	1	0.07	0.16



32	3540745	BODY, CHECK VALVE HQ/PQ	1	0.23	0.51
33	62374	BALL, 22MM	1	3.28	7.22
34	100865	PIN, SPRING 3/16 X 1-3/4 LG	1	0.05	0.11
35	3540737	BUSHING, HQ/PQ Q/D VALVE	1	0.02	0.04
36	3543804	INNER-TUBE CAP	1	2.76	6.07

PQ HEAD OPTIONS					
18A	3543378	FLUID RETENTION BUSHING, HEAVY	1	0.03	0.07
19A	104971	FLUID RETENTION SPRING, MEDIUM	1	0.05	0.11
19B	104972	FLUID RETENTION SPRING, HEAVY	1	0.05	0.11
25A	3540300	WASHER SHUTOFF VALVE	4	0.00	0.00
A	5001333	DRY HOLE VALVE KIT HQ		2.32	5.11
16	3548544	VALVE PISTON	1	0.08	0.18
17	306131	FLUID RETENTION BUSHING	1	0.03	0.07
17A	3543378	FLUID RETENTION BUSHING, HEAVY	1	0.03	0.07
18	104966	FLUID RETENTION SPRING, LIGHT	1	0.03	0.07
18A	104971	FLUID RETENTION SPRING, MEDIUM	1	0.05	0.11
19	3548715	PIN, SPRING 1/4" X 2 3/4"	1	0.03	0.07
20	3542987	HEAD ASSEMBLY TOOL (NOT SHOWN)	1	0.23	0.51
21	3543407	KIT BAG (NOT SHOWN)	1	0.45	1.00

Boart Longyear – Overshot Product



OVERSHOTS

OVERVIEW

Boart Longyear's overshots are an integral part of the Q[®] Wireline system. Once the inner tube is full of core, the overshot is lowered (or pumped-in for an inclined hole) to the bottom of the hole. Its heavy-duty lifting dogs grab a secure hold at the top of the inner tube assembly.

As part of Boart Longyear's commitment to safety on the drill site, we offer the patented EZY-Lock overshot for use with the Q[®] Wireline system on surface applications.

For maximum productivity in underground or pump-in applications, Boart Longyear offers the new Quick Pump-In overshot.

The new Quick Pump-In overshot provides significantly improved pump-in rates with dual pump-in lip-seals. In addition, it incorporates the proven/reliable, patented Landing Indicator bushing as an optional bypass valve feature. When drilling flat or slightly declined, the landing indicator bushing and valve ball (or valve piston in AQTK) will provide a pressure spike upon landing and allow fluid to bypass the overshot when pulling from the hole, significantly improving retraction rates and eliminating the need to refill the hole with drilling fluid.

CONVERSION KITS

Boart Longyear has set up conversion kits to assist our customers in converting over, instead of the expense of buying a completely new system. There are also kits available to convert between surface and underground applications.

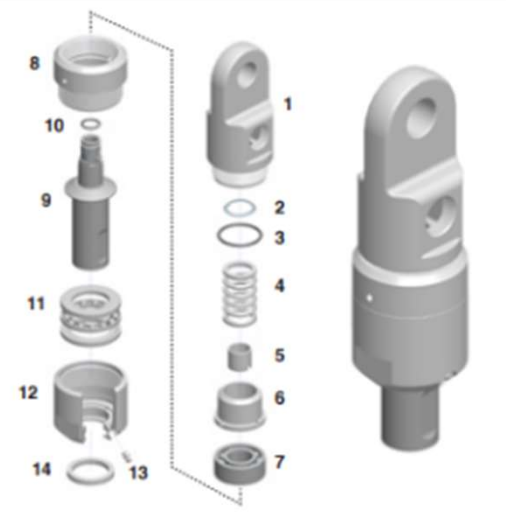
Boart Longyear – Compact Water Swivel Product

3546952 - COMPACT PLUS WATER SWIVEL

ID #	PART #	DESCRIPTION	KG	LB
	3546952	COMPACT PLUS WATER SWIVEL	7.6	16.7
			QTY	
1	3546529	SWIVEL CAP	1	
2	23863	WAVY WASHER	1	
3	19678	O RING	1	
4	23861	SWIVEL PACKING	1	
5	3546530	SLOTTED WEAR SLEEVE	1	
6	23866	PACKING CAGE	1	
7	20324	THRUST BALL BEARING	1	
8	3546531	UPPER SWIVEL BODY	1	
9	3546907	SPINDLE	1	
10	20169	O RING	1	
11	3546908	THRUST BALL BEARING	1	
12	3546909	LOWER SWIVEL BODY	1	
13	17447	GREASE FITTING	1	
14	25612	FELT SEAL	1	

Features:

- #16 NPT hose connection.
- AW rod box spindle connection.

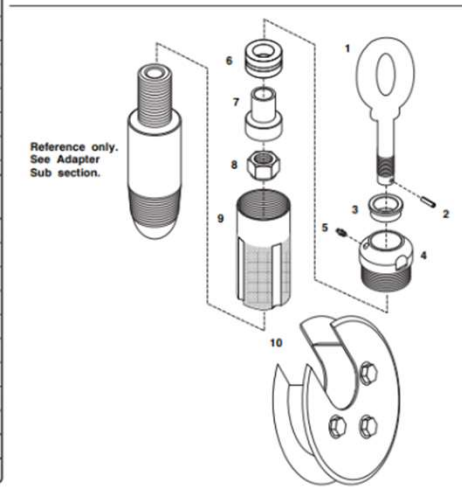


Boart Longyear –Hoisting Plug Product

Features:

- 12.5 mm (0.5 in) diameter down-angled vent is incorporated into the body to eliminate side spray from pressurized drilling fluid.
- Long-life thrust roller bearing provides pivot for ease of connecting and reduces hoisting cable winding during hoisting and lowering.
- NW rod box spindle connection.
- Optional shackle guide is recommended to eliminate jamming of certain hoist cable shackles in the eye bolt.

ID#	PART #	DESCRIPTION	QTY
	3545380	HOISTING PLUG - STANDARD	
1	3545376	EYE BOLT	1
2	37394	SPRING PIN 1/4" X 1-3/4"	1
3	3545377	WEAR SLEEVE	1
4	3545378	BEARING HOUSING	1
5	17447	GREASE FITTING	1
6	3542992	HOIST PLUG BEARING	1
7	3545379	SHAFT SLEEVE	1
8	3543012	STOVER LOCK NUT	1
9	3545381	LOWER HOUSING	1
	3545380	HOISTING PLUG HEAVY DUTY	
1	3542899	EYE BOLT	1
2	24307	SPRING PIN 1/4" X 2"	1
3	3542900	WEAR SLEEVE	1
4	3542901	BEARING HOUSING	1
5	17447	GREASE FITTING	1
6	3542902	THRUST BEARING	1
7	3542903	SHAFT SLEEVE	1
8	3542914	STOVER LOCK NUT	1
9	3542904	LOWER HOUSING	1
OPTIONS			
10	3542907	SHACKLE GUIDE	1



Boart Longyear –Casing Cutter Product

CASING CUTTERS

Operations:

Upon lowering the assembly, the cutting lugs are retracted inside the body by the spring. With low rotation (25 rpm) initially, the cutting lugs are forced out using water pressure, 1,360 kpa (200 psi). A drop in pressure indicates that the cutting operation is complete.

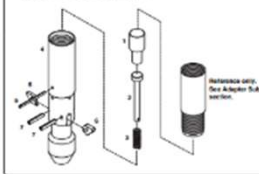
ID #	PART #	DESCRIPTION	QTY
1-10	354305	AW CASING/8" ROD CUTTER ASSEMBLY	
1	30816	PISTON	1
2	60150	PLUNGER	1
3	26480	SPRING	1
4	354307	BODY	1
5	25069	SPRING PIN 1/8" X 1-3/4"	1
6	60156	CASING CUTTER LUG	2
7	23175	SPRING PIN 1/4" X 1"	2

1-10	52775	BW CASING CUTTER ASSEMBLY	
1	51255	RETAINING RING	1
2	52776	PLUNGER CASE	1
3	21837	STAINLESS STEEL BALL 15/16"	1
4	14520	WATER TUBE	1
5	14519	PLUNGER	1
6	15280	SHEAR PIN	1
7	14522	SPRING	1
8	14521	SPRING REST	1
9	14517	BODY	1
10	14523	CASING CUTTER LUG	2
11	14587	CASING CUTTER PIN	2
12	44863	SET SCREW	1

1-10	52777	NW CASING/7 1/4" ROD CUTTER ASSEMBLY	
1	61460	RETAINING RING	1
2	52776	PLUNGER CASE	1
3	26167	STAINLESS STEEL BALL 1-5/8"	1
4	14529	WATER TUBE	1
5	14528	PLUNGER	1
6	22934	PIN 3/16" X 1-1 1/16"	1
7	14531	SPRING	1
8	14530	SPRING REST	1
9	14526	BODY	1
10	14532	CASING CUTTER LUG	2
11	14588	CASING CUTTER PIN	2
12	44863	SET SCREW	1

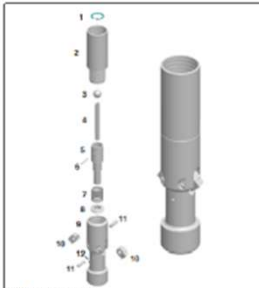
AW Applications:

- Used for cutting AW casing or "B" rod at any point in a drill hole when the entire string can not be retrieved.
- ARQTK rod box connection.



BW Applications:

- Used for cutting BW casing at any point in a drill hole when the entire string can not be retrieved.
- BQ rod box connection.



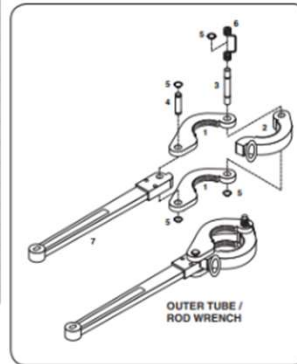
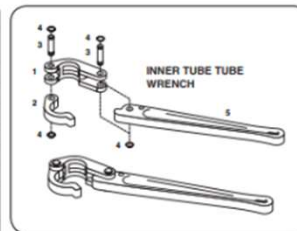
NW Applications:

- Used for cutting NW casing or HIRQ/HQHMQ rod at any point in a drill hole when the entire string can not be retrieved.
- NQ rod box connection.

Boart Longyear –Coring Wrenches Product

CORING WRENCHES - PQ

ID #	PART #	DESCRIPTION	QTY	KG	LB
1-5	26171	FULL GRIP PQ INNER TUBE WRENCH		3.6	7.9
1	28972	STATIONARY JAW	1	0.52	1.14
2	28973	SWING JAW	1	0.77	1.69
3	38580	PIN	2	0.18	0.40
4	38583	RETAINING RING	4	0.40	0.88
5	28259	WRENCH HANDLE	1	1.56	3.43
1-7	45804	FULL GRIP PQ OUTER TUBE WRENCH		11.02	24.29
1	45989	STATIONARY JAW	2	1.90	4.18
2	45988	SWING JAW	1	1.93	4.25
3	43484	PIN, 5/8" X 4"	1	0.16	0.35
4	43485	PIN, 5/8" X 2-3/4"	1	0.11	0.24
5	38583	RETAINING RING	4	0.40	0.88
6	43422	SPRING	1	0.02	0.04
7	43483	WRENCH HANDLE	1	3.40	7.48
1-7	45792	FULL GRIP H ROD WRENCH		7.6	16.7
1	45987	STATIONARY JAW	2	2.00	4.40
2	45986	SWING JAW	1	1.86	4.09
3	43484	PIN, 5/8" X 4"	1	0.16	0.35
4	43485	PIN, 5/8" X 2-3/4"	1	0.11	0.24
5	38583	RETAINING RING	4	0.40	0.88
6	43422	SPRING	1	0.02	0.04
7	43483	WRENCH HANDLE	1	3.40	7.48



Boart Longyear – Drilling Best Practice

Pada umumnya, pengeboran slim hole / coring menggunakan pipa diameter kecil dari mulai NQ, HQ, PQ dan PW casing. Setiap pipa hanya punya anulus 1 inch. Dan hanya memerlukan sedikit air dengan menggunakan pompa pressure FMC 435 2-inch triplex piston. Kekentalan air hanya diperlukan sekitar 30 – 40 ppm, dengan debit air paling banyak 50 liter per menit untuk pipa ukuran PQ (diameter 4,5 inch).

Core barrel adalah bagian pipa paling depan dalam rangkaian pengeboran, terdiri dari bits, reamers, outer tubs, adapter coupling dan locking coupling. Didalamnya, terdapat inner tube yang nantinya inner tube tersebut akan berisi batuan yang dibor.

Pada saat batuan sudah penuh atau ada indikasi batuan tidak dapat masuk kedalam inner tube, itu saat yang tepat mengangkat inner tube menggunakan overshot dan mengisi kembali core barrel dengan inner tube yang kosong dan melanjutkan kembali pengeboran. Kekentalan cairan tergantung pada saat batuan yang sedang dibor. Semakin rusak batuan, berarti kekentalan akan makin tinggi. Batuan yang press atau compact, kekentalan cairan pengeboran bisa lebih rendah.

Boart Longyear – World Drilling Magazine

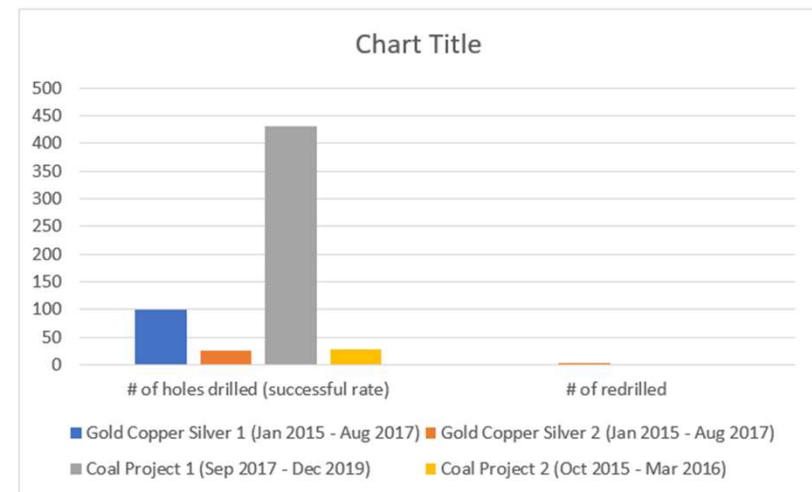
Top exploration drilling
contractors for 2021 –
Statistics of diamond
drilled meters

Boart Longyear – World Drilling Magazine



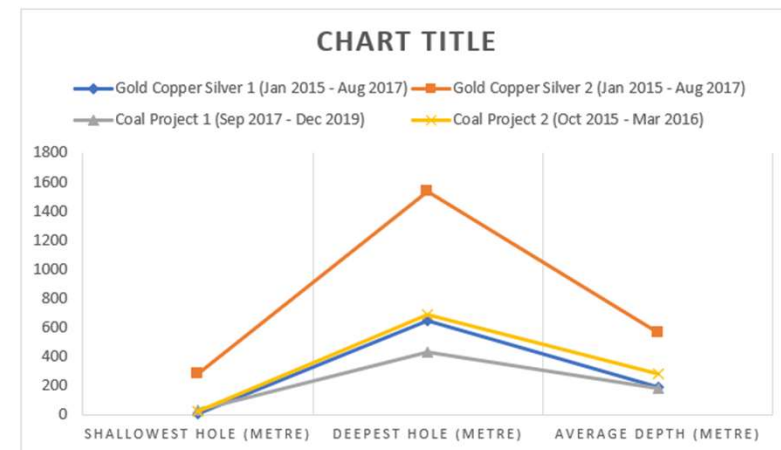
Boart Longyear – Indonesia Drilling Statistic

Project	Drilling	
	# of holes drilled (successful rate)	# of redrilled
Gold Copper Silver 1 (Jan 2015 - Aug 2017)	100	0
Gold Copper Silver 2 (Jan 2015 - Aug 2017)	25	2
Coal Project 1 (Sep 2017 - Dec 2019)	432	0
Coal Project 2 (Oct 2015 - Mar 2016)	27	0



Boart Longyear – Indonesia Drilling Statistic

Project	Depth			
	Shallowest hole (metre)	Deepest hole (metre)	Average depth (metre)	
Gold Copper Silver 1 (Jan 2015 - Aug 2017)	10	650.3	188.37	HQ3
Gold Copper Silver 2 (Jan 2015 - Aug 2017)	287	1532.1	567.13	BQ3
Coal Project 1 (Sep 2017 - Dec 2019)	36	436.5	184	HQ3
Coal Project 2 (Oct 2015 - Mar 2016)	27.4	686.6	281.19	HQ3



Slim – core hole Improvement Plan

- Safe Operation
- Time Effective
- Cost efficient



Well Objective - Scale of Priority



Well design – Optimize a coring



Well control – a conservative risk assumption and adequate mitigation



Rig and Equipment selection – An existing equipment optimization



Rig contractor selection – proven experience and track record



Thank you

